Home Network & Media Wall Plan

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The plan

One plan for the whole house: a three-unit Deco mesh wired by Cat6, and a built-in media wall in the 225 × 220 cm living room alcove, set in a room painted top to bottom in Farrow & Ball Card Room Green. The Virgin Hub 5 becomes a plain modem and the main Deco takes over as router. The PC and Series X plug straight into the LG C6, and every other console shares one input through an HDMI switch. A RetroTINK upscaler makes PS2 games look their best on the 55-inch screen. Every LP and most singles are face-out, in two pull-out trays, a browser under the turntable and a ledge above the TV, and games live in a roll-out drawer with every title visible. It all costs about £2,086–4,082 plus the electrician, tracked item by item below. Above the worktop, solid oak uprights carry the shelves on joints left on show, with warm light from the ceiling strip, the uprights and a paper lantern; the slat wall is kept as the alternative.

Recommendations at a glance

Area	Recommendation	Why
Internet	Virgin Hub 5 in modem mode; Deco A becomes the router	One router: Open NAT for the Xbox, and HomeShield and parental controls work
Mesh	One Deco per floor, each wired back to an 8-port Switch 1 by Cat6	Full speed on every floor; dual-band Decos slow down over wireless hops
Living room network	8-port switch wiring the Series X, Xbox One, Xbox 360, PS3, PC and TV	Lowest lag for games, phones get the Wi-Fi to themselves, one port spare
Room colour	Colour-drench the room in Farrow & Ball Card Room Green No. 79: matt walls and ceiling, the same colour in Estate Eggshell on the skirting (now white), door and trunking	The media wall becomes part of the room; a soft grey-green that lets the oak and the records lead
Lighting	Keep the LED strip round the ceiling (upgrade to a diffused channel later), LED grooves in the oak uprights, and a four-head cone spotlight bar on the existing ceiling rose, all 2700K and dimmable	Warm, layered light that suits the hand-made oak and keeps glare off the OLED
TV inputs	HDMI 1 PC, HDMI 2 soundbar (eARC), HDMI 3 Series X, HDMI 4 five-way switch	PC and Series X keep 120Hz+ and VRR; every other console shares one input
PS2 and PS1	RetroTINK-6X CE: PS2 by component and optical, PS1 by RGB SCART, powered with the retro consoles	Motion-adaptive deinterlacing keeps PS2 games clean on a 55" OLED; optical carries 5.1 where games have it
Vinyl	New stylus; turntable → phono preamp (on a shelf over the sub) → soundbar AUX, in Music mode; about 230 LPs, every one face-out, and about 310 singles: two thinner trays in bay 3, a browser under the deck and a ledge above the TV	Biggest sound gain for the money, and the collection is on display right where you play it

Media wall	Artisan oak: two solid oak uprights with pegged and through-tenoned shelves above a solid oak worktop at 76 cm; below, two BESTÅ frames on a 9 cm plinth, a ply bay for the records and an open sub bay behind one oak face frame. Xbox bay left with a roll-out game drawer, PlayStation bay centre, vinyl bay right with two face-out record drawers	Every console fits even at its largest version, each zone sits under the kit it belongs to, and every game title and record cover is visible
Sound	Sub in the alcove's right corner facing the sofa; soundbar centred; rears either side of the sofa, linked by a cable behind it	A corner gives the most bass; the surround is built around the seats opposite the TV
Power	The low socket on the side wall in front of bay 1 becomes a double: your 10-gang behind the TV (side A left on, side B the retro switch) and a 6-way with a switch per socket, which keeps Deco B always on	Retro gear fully off between sessions; the TV stays in standby for OLED care
PC	Oak wall shelf on the left wall, glass to the room, back screened by an oak frame with woven cane, keyboard tray underneath; active optical HDMI cable for the 4–5 m route	On show without losing 4K at 120Hz+, and no cable mess facing the sofa
Sound	Every source plays through the Hisense: HDMI sources via the TV's eARC, vinyl, PS2 and phones straight in	One volume control and full 5.1 wherever a source has it

Build tracker

Every purchase and job in build order comes to about £2,086–4,082, plus the electrician's quote. Tick items off as you go and write in what you paid. Estimates are approximate UK prices. New items are marked.

Phase	Estimate (£)	Biggest item
Network	51–70	two 8-port switches
Cabling and power	130–230, plus electrician	Cat6 cable and trunking
Decorating	220–540	paint
Media wall	567–1,117	record trays and the bay 3 carcass
TV and consoles	283–410	RetroTINK-6X CE
Vinyl	70–165	phono preamp
Oak joinery and lighting	765–1,550	solid oak worktop and shelves
Total	**2,086–4,082, plus electrician**	

Already owned: Virgin Hub 5, three Deco X1500s, two TL-SG105S switches (both become spares), LG OLED C6 55" with its stand, a 10-gang extension with a switch per five-socket side, Hisense AX5100G-class 5.1 soundbar, the Series X fan stand, the slim turntable and all the consoles.

Status	Item	Phase	Qty	Estimate (£)	Paid (£)
☐ To do	Put the Virgin Hub 5 into modem mode	Network			
☐ To do	Set up Deco A, Switch 1 and both PCs	Network			
☐ To do	Add Deco B and C; speed-test each room and the Xbox	Network			
☐ To buy	Cat6 patch leads, 0.5–2 m	Network	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port upstairs **(new)**	Network	2	36–50	
☐ To do	Book one electrician visit: both Cat6 runs plus a spare each (Deco B's through the ceiling in the alcove's right corner), swap the single socket on the side wall in front of bay 1 for a double (or move it 40 cm back into the alcove, behind the unit) **(changed)**, a socket by the sofa for the rear speakers if there isn't one, a socket under the PC shelf, a switched fused spur for the LED drivers, and an LED-compatible dimmer for the ceiling light **(new)**	Cabling and power		Quote	
☐ To buy	Cat6 solid pure copper, round (not CCA)	Cabling and power	2 runs	30–50	
☐ To buy	6-way surge-protected extension with individually switched sockets, for Deco B, the 8-port switch, turntable, preamp and RetroTINK (your 10-gang takes the rest)	Cabling and power	1	25–40	
☐ To buy	4-port USB charger, 5V 2A per port (HDMI switch, Xbox fan stand, controller dock)	Cabling and power	1	10–20	
☐ To buy	USB plug, 5V 2A or more, for the RetroTINK on the switched strip **(new)**	Cabling and power	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels, corner trunking	Cabling and power	1	20–30	
☐ To buy	Two-compartment trunking for the service channel, about 2 m	Cabling and power	1	10–20	
☐ To buy	Slim paintable trunking for the PC drop and skirting run, about 3 m	Cabling and power	1	15–30	
☐ To buy	DC extension lead for the Deco adapter, if its lead won't reach down the upright's groove to the 6-way (match the plug)	Cabling and power	1	5–10	
☐ To buy	PC tidy kit: braided sleeve, right-angle C13 power lead, shelf-notch grommet **(new)**	Cabling and power	1	10–20	
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check by day and with the LED strip on **(changed)**	Decorating			
☐ To buy	Farrow & Ball Card Room Green No. 79: Estate (or Modern) Emulsion for walls, ceiling and alcove; Estate Eggshell for skirting, door, frame and trunking; primer/undercoat **(changed)**	Decorating		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door **(new)**	Decorating	1	20–40	
☐ To do	Paint the whole room, alcove included, before the unit goes in; skirting and door from white to Card Room Green eggshell	Decorating			
☑ Done	Check the alcove back wall for the shelf fixings (solid)	Media wall			
☐ To do	Check the left wall where the PC shelf goes: solid, or plasterboard on studs **(new)**	Media wall			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2 **(changed: bay 3 is now custom)**	Media wall	2	70–200	

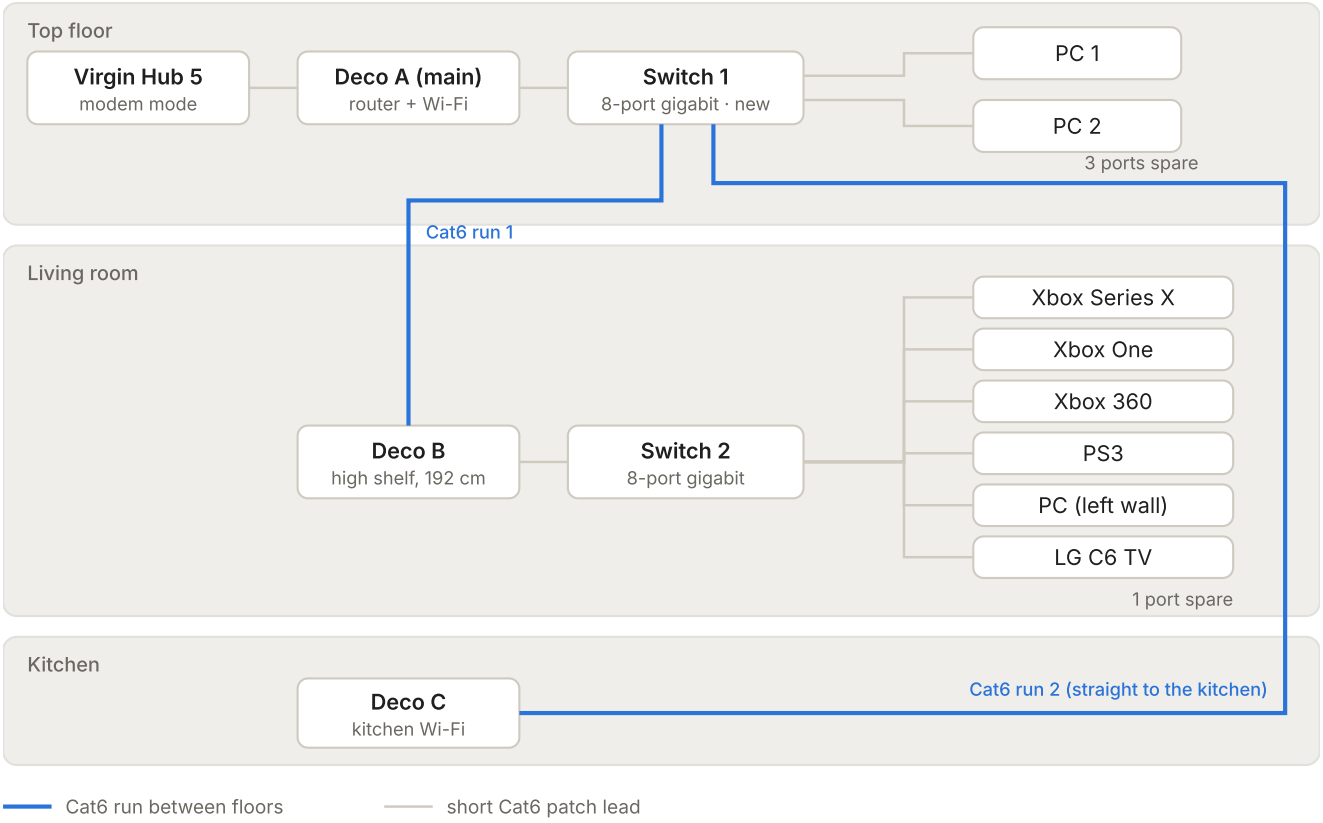
Status	Item	Phase	Qty	Estimate (£)	Paid (£)
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm, plus shims **(new)**	Media wall	1	15–30	
☐ To buy	Bay 3 carcass: 18 mm birch ply cut to size (sides 73 × 40, top and bottom 56.4 × 40, two back rails), connector bolts **(new)**	Media wall	1	60–110	
☐ To buy	Felt-lined battens for the worktop over the sub bay	Media wall	1	5–10	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	Media wall	4	12–32	
☐ To buy	Anti-tip wall brackets and 10 cm spacer blocks for the two frames and the bay 3 carcass	Media wall	1 set	15–25	
☐ To buy	Two face-out record tray kits: 12 mm and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg **(changed: thinner, singles face-out)**	Media wall	2	150–240	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer **(new)**	Media wall			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt **(new)**	Media wall	1	50–90	
☐ To buy	Hidden cleats for the shelf ends on the side walls, 44 × 22 mm planed softwood, about 3 m	Media wall	1	15–25	
☐ To buy	Bookends for the high shelf **(new)**	Media wall	2	10–20	
☐ To buy	Lidded box for cables and controllers on the high shelf	Media wall	1	10–20	
☐ To buy	LP browser for the worktop under the deck: oak base, low front rail and side, 44 × 24 cm	Media wall	1	30–60	
☐ To buy	Kit shelf over the sub: oak or ply 44 × 30 cm on battens, top at 38.8 cm, with rubber pads	Media wall	1	15–30	
☐ To buy	Hidden slim drawer under the worktop in the sub bay: oak front 44 × 7.4 cm, 12 mm ply box 40 × 28 × 6 cm, 250 mm full-extension runners **(new)**	Media wall	1	25–45	
☐ To buy	PC shelf rebuilt as a box with a keyboard and mouse tray: oak, 300 mm light runners, French cleat **(new)**	Media wall	1	25–50	
☐ To buy	Oak for the PC shelf box: 27 mm top 65 × 32, 18 mm sides; French cleat and 2 steel brackets rated 25 kg+	Media wall	1	30–70	
☐ To buy	Compact isolation riser for the sub (about 30 × 35 cm, 4–6 cm tall), plus isolation feet for the turntable	Media wall	1	30–60	
☐ To do	Build the plinth and bay 3 carcass, fix everything to the wall; set rows to the fit check heights; fit the game drawer and both record trays; place the TV, soundbar, Series X and sub	Media wall			
☐ To buy	5-in-1 4K60 HDMI switch, auto-switching, with remote	TV and consoles	1	20–35	
☐ To do	Measure the PC-to-TV cable route (estimated 4–5 m)	TV and consoles			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5–7.5 m	TV and consoles	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m	TV and consoles	5	20–30	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	TV and consoles	1	200–240	
☐ To buy	PS2 component video cable, good quality	TV and consoles	1	10–20	

Status	Item	Phase	Qty	Estimate (£)	Paid (£)
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	TV and consoles	1	5–10	
☐ To buy	RGB SCART cable for the PS1	TV and consoles	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK	TV and consoles			
☐ To buy	Replacement stylus for the slim turntable	Vinyl	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	Vinyl	1	40–100	
☐ To buy	RCA-to-3.5 mm lead	Vinyl	1	5–10	
☐ To buy	Carbon-fibre record brush	Vinyl	1	10–15	
☐ To do	Set tracking weight and soundbar Music mode	Vinyl			
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off	Finishing			
☐ To do	Deco app: prioritise Series X and PC, guest Wi-Fi, auto firmware	Finishing			
☐ To do	Label each switched socket and HDMI switch input with the console's name	Finishing			
☐ To do	Set the sub and rear speaker levels from the sofa	Finishing			
☐ To buy	Two solid oak uprights, 150 × 30 × 4.5 cm, quarter-sawn if possible **(new: main plan)**	Oak joinery and lighting	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in three lengths (128.9, 45.7, 46.2 × 30 × 4), left shelf 51 × 25 × 4, turntable shelf 51.5 × 45 × 4, record ledge 128.9 × 7 × 2.4	Oak joinery and lighting	1 set	200–350	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets	Oak joinery and lighting	1	15–30	
☐ To buy	Solid oak worktop, full stave, cut to 225 × 40 × 3 cm, chamfered edge (the offcut makes the LP browser base)	Oak joinery and lighting	1	200–400	
☐ To buy	Oak face frame, 20 mm, stiles and rails	Oak joinery and lighting	1	80–180	
☐ To buy	Cane screen for the PC: oak frame and pre-woven cane, 26 × 48 cm	Oak joinery and lighting	1	20–40	
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, driver; 6500K bias light for the TV	Oak joinery and lighting	1	40–90	
☐ To buy	Ceiling light: Dunelm Elements Wolston 4-light spotlight bar (£49), in the multicoloured style, plus four warm-white (2700K) dimmable E14 LED bulbs, 6 W max **(changed)**	Oak joinery and lighting	1	60–80	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	Oak joinery and lighting	1	0–80	
☐ To do	Cut the joints, fit the uprights, shelves, ledge and face frame, notch the uprights for the ceiling strip, and wire the LEDs (see the build guide)	Oak joinery and lighting			

Network

Every floor has its own Cat6 link back to the top floor

Revised: an 8-port switch upstairs leaves room to grow, and the Xbox 360 is wired too

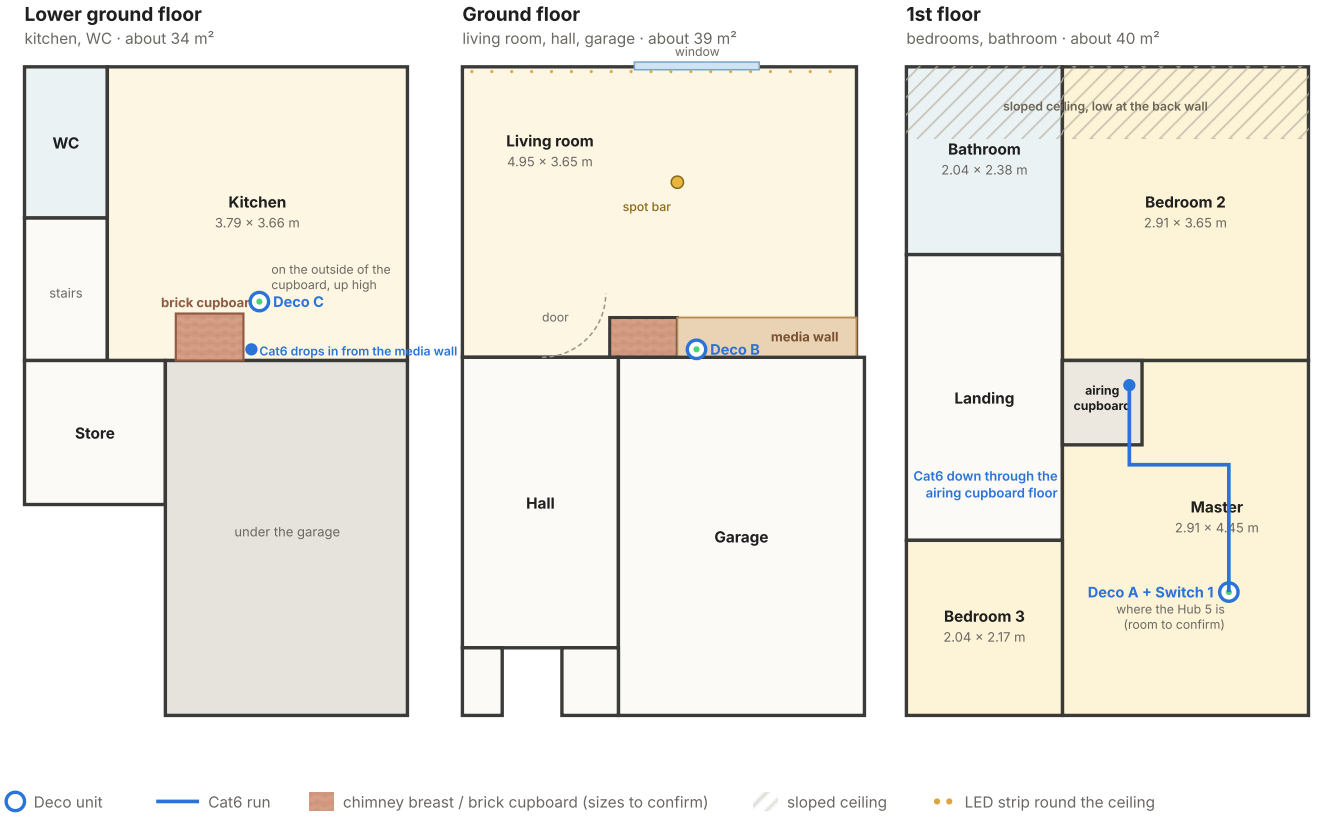


The hub feeds Deco A, and Switch 1 sends a Cat6 run down to each of the other Decos. In the living room, Deco B feeds the 8-port switch, which gives every networked console, the PC and the TV its own cable. Each Deco X1500 has two gigabit ports, so one takes the incoming cable and the other feeds the switch.

Around the house

The house: where the Decos and Cat6 runs go

Each floor drawn to the same scale. Both Cat6 runs drop through the airing cupboard, which sits right over the media wall's corner



Traced from the agent's plan (north, the window side, at the top), which leaves out the alcoves, the chimney breast, the brick cupboard and the slopes; those are added here from your notes and photos.

- **Deco A and Switch 1** stay upstairs with the Hub 5 and both PCs. Both Cat6 runs go to the airing cupboard on the landing, which sits right over the media wall's right-hand corner, and drop through its floor, often beside the existing pipes.
- **Run 1 to Deco B** comes down through the living room ceiling in the alcove's right corner, straight to Deco B on the high shelf. Only a short length shows, between the ceiling and the shelf.
- **Run 2 to Deco C** follows run 1 down the groove behind the right upright to the cable channel, then through the floor into the kitchen below, where the brick cupboard sits under the chimney breast. Fix Deco C on the outside of the cupboard, up high on the kitchen side: inside the brick it would lose much of its Wi-Fi.
- **Lengths:** roughly 6–9 m for run 1 and 9–13 m for run 2, depending on where the Hub 5 is; measure the real route and add 2–3 m to each.
- **Sloped ceilings:** the bathroom and bedroom 2 slope down towards the back wall. Keep any Deco, switch or desk in bedroom 2 on the high side, and run cables along the floor there, not the ceiling.

What each part does

Part	Job	Why it's needed
Virgin Hub 5, modem mode	Brings the internet into the house, nothing else	Stops two routers fighting (double NAT), which breaks Xbox online play. Its Wi-Fi switches off, so it no longer clutters the airwaves.
Deco A, top floor (main)	The router: runs the network, Wi-Fi, parental controls and HomeShield	Everything starts here. HomeShield and parental controls only work when the Deco is the router.

Switch 1, top floor, 8-port	Turns Deco A's one spare port into seven	Feeds both PCs plus the two cables down to the other floors, with three ports left for later.
Deco B, living room	Wi-Fi for the middle floor, and the wired point for the TV area	Fixes the 5GHz dead side of the living room.
Switch 2, living room, 8-port	Turns Deco B's spare port into seven	The Series X, Xbox One, Xbox 360, PS3, PC and TV each get a cable, with one port spare.
Deco C, kitchen	Wi-Fi for the bottom floor	The kitchen currently gets poor signal on every band.
Cat6 runs between floors	Wired link between Decos ("Ethernet backhaul")	Without it, Decos relay each other over the same 5GHz phones use, losing speed at each hop.

How to set it up

The build runs in two stages: get the mesh working now, then add the cables between floors when they're run.

Stage 1: mesh up today (no cables between floors yet)

1. Unplug everything from the hub. Log in at 192.168.0.1 with the password on its base, then choose Super Hub → Modem Mode → Enable Modem Mode → Apply Changes. Afterwards its settings page moves to 192.168.100.1.
2. Connect hub port 1 to Deco A and set it up in the Deco app, reusing the old Wi-Fi name and password. Virgin's guide doesn't name the port, so if the Deco shows no internet, try another port and restart the hub.
3. Plug the new 8-port Switch 1 into Deco A's other port, then both PCs into Switch 1.
4. Add Deco B on the living room's high shelf. Connect the 8-port switch to it, then the Series X, Xbox One, Xbox 360, PS3, PC and TV.
5. Add Deco C in the kitchen, on the side nearest the living room. For now it links over Wi-Fi.
6. Check the Xbox network settings show NAT: Open.

Stage 2: add the Cat6 runs

1. Run Cat6 from Switch 1 to Deco B, and a second run from Switch 1 straight to Deco C.
2. Plug each run into the Deco's free port. On Deco B, Switch 2 stays in the other port.
3. In the Deco app, confirm Deco B and Deco C show as "Ethernet".
4. Move each Deco to the centre of its floor for best coverage. They no longer need to see each other.
5. In the Deco app, give the Series X and PC high priority under QoS.

Cabling rules

- **Cable type:** solid pure-copper Cat6, round. Avoid "CCA" (copper-clad aluminium) and flat cable for long runs.
- **Length:** measure the actual route, not the straight line, and add 2–3 m spare. Anything under 100 m runs at full gigabit.
- **Routing:** go through the floor beside pipes, along skirting, or down an outside wall with outdoor-rated cable. Cross mains cables at right angles rather than running alongside them.
- **Separation:** in the media wall's service channel, mains goes in one compartment of the trunking and HDMI, Cat6 and the turntable leads in the other.
- **Handling:** no kinks or tight folds, never staple through the cable, and label both ends.
- **No loops:** never connect two cables between the same pair of devices. A loop can take the whole network down.

Done when

- Deco app shows Deco B and Deco C as "Ethernet"
- Each PC's network status shows 1.0 Gbps (100 Mbps means a bad cable: swap it)
- Xbox shows NAT: Open
- Phones hold a strong signal in the kitchen and on both sides of the living room

Media wall proposal

The main plan is now the artisan oak build: oak used as furniture and joinery against a wall painted in Farrow & Ball Card Room Green. Below the worktop is one full-width unit: bays 1 and 2 are IKEA BESTÅ 60×40×64 frames on a 9 cm plinth, bay 3 is a custom plywood carcass with two face-out record trays, and the 45 cm bay at the right stays open for the subwoofer, all behind one oak face frame under a 3 cm solid oak worktop at 76 cm. Above it, two solid oak uprights frame the TV and carry the shelves on joints left on show: wedged through-tenons, pegged housings and a record ledge above the screen. Each bay holds one family: Xbox under the Series X, PlayStation under the TV, vinyl under the turntable. Every console fits even at its largest version, every record and single is face-out, and every cable drops into a hidden channel.



The artisan oak media wall in Farrow & Ball Card Room Green, drawn to the real measurements in a one-point perspective, with the game drawer and the lower record tray rolled out 30 cm and an inset of each from above. The view now takes in the ceiling: the existing LED strip round the room's edge and the new spotlight bar on the existing rose. Dashed lines show what's hidden: gold for Deco B's leads and the 10-gang, coral for mains from the double socket on the side wall. The numbers match the key below.

OAK JOINERY	
1	Solid oak uprights, worktop to ceiling, 4.5 × 30
2	Wedged through-tenons: left shelf, top 156
3	Wedged through-tenons: turntable shelf, top 113
4	Oak pegs where the shelves are housed
5	High shelf, top 192: books up to 26 cm
6	Record ledge above the TV: 4 LPs face-out
7	LED groove in each upright washes the wall
ON THE WORKTOP	
8	LG OLED C6 55in on its stand · centre 116.5
9	Soundbar · eARC, optical and AUX inputs
10	Xbox Series X on its fan stand · 37 tall
11	Turntable on its oak shelf · 40 cm lid clearance
12	LP browser under the deck · ~50 LPs face-out
13	Solid oak worktop 225 × 40 × 3, top at 76
IN THE BASE	
14	Oak face frame; its bottom rail hides the 9 cm plinth
15	Xbox One + power brick · bay 1 top row
16	Xbox 360 + power brick · bay 1 middle row
17	Game drawer: 2 face-out bins, spine-up rows behind
18	PS3 + controller basket · bay 2 top row
19	PS2 + HDMI switch · bay 2 middle row
20	PS1 + RetroTINK · bay 2 bottom row
21	Upper record tray: ~90 LPs + ~120 singles, all face-out
22	Lower record tray: the same, shown pulled out
23	Kit shelf: preamp, brush, ~70 singles
24	Subwoofer on a 5 cm riser
25	Hidden slim drawer under the worktop

POWER, NETWORK AND LIGHT	
26	Deco B, powered: leads in the groove behind the upright
27	Cat6 from the top floor, through the ceiling
28	Double socket on the side wall, in front of bay 1
29	10-gang extension, hidden behind the TV
30	6-way switched strip: Deco, switch, deck, preamp
31	LED strip round the ceiling edge (as now)
32	Ceiling light: 4-spot cone bar on the existing rose
PC CORNER	
33	PC, glass side to the room · shelf top 110
34	Oak frame with woven cane hides the PC's ports
35	Keyboard + mouse tray, rolls out 25 cm
36	PC cables: shelf notch → trunking → skirting



The same view with every drawer closed. The front covers of the two game bins, the front LP and single in each record tray, the browser's front record and the four LPs on the ledge stay on show. Full-resolution copies of both renders come with this plan.

Built to your alcove: 225 cm wide, 50 cm deep, floor to ceiling (220 cm)

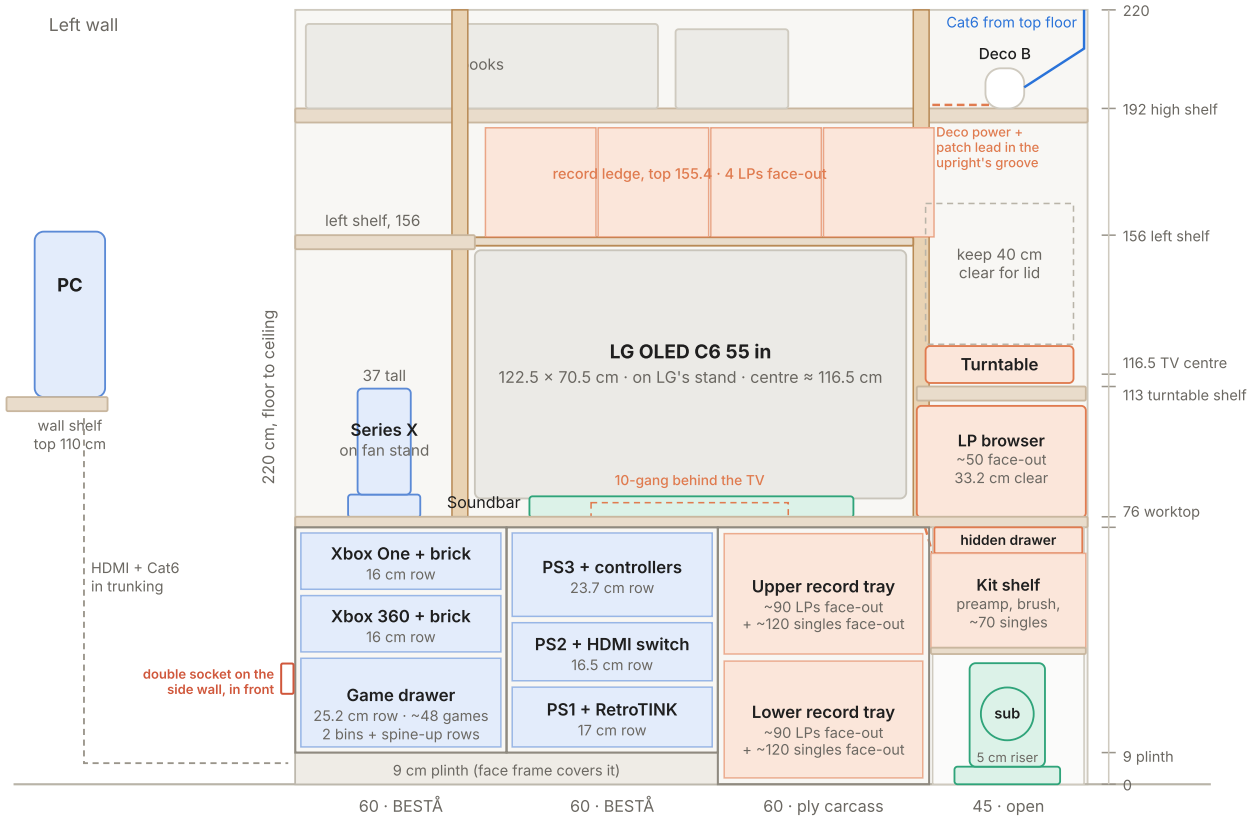
- **Bays 1 and 2:** two BESTÅ frames, each 60 × 40 × 64 cm, stand on a 9 cm plywood plinth, so their tops are at 73 cm. Back panels off. The oak face frame's bottom rail covers the plinth, so it never shows.
- **Bay 3:** a custom carcass in 18 mm birch ply, 60 × 40 × 73 cm, standing on the floor with its top level with the frames. Inside it is 56.4 cm wide and 69.4 cm tall: exactly two face-out tiers of 34.7 cm. The drawer runners screw straight into the ply sides, so it needs no liners.
- **Fixing:** the frames and carcass are bolted to each other, stand 10 cm off the back wall on spacer blocks, and are fixed to it with anti-tip brackets. With two trays of about 32 kg each, the anti-tip brackets on bay 3 are essential, and only one record tray should be open at a time.
- **Worktop:** solid oak, full stave or wide board, 225 × 40 × 3 cm with a chamfered front edge, flush with the fronts, top at 76 cm. The 10 cm left open behind it is the slot for cables and rising warm air.
- **Oak uprights:** two, 4.5 × 30 cm, from the worktop to the ceiling at 44.5–49 and 175.5–180 cm from the left wall, about 2 cm clear of the TV each side. They carry the high shelf, the record ledge, the left shelf and the turntable shelf (see "The oak joinery").
- **Power:** everything runs from the wall socket on the side wall just in front of bay 1, about 30 cm up, swapped from a single to a double: one socket for your 10-gang behind the TV, one for the 6-way at the back of the sub bay. The two leads drop to the floor, enter the unit through a small notch at the left end of the face frame's bottom rail, run back through the plinth and along the mains side of the channel. If the electrician is changing the socket anyway, moving it about 40 cm back into the alcove, behind bay 1, hides it and the plugs completely.
- **Inside bays 1 and 2:** about 56.8 cm wide, 38–40 cm deep with the back open, and 60.8 cm high, exactly as before: every row and console just sits 9 cm higher. The game drawer's base is now 11.6 cm off the

floor, easier to reach.

- **Matching fronts:** the game drawer and both record trays have oak lips with a hand-cut finger pull (6.5 cm on the game drawer, a slimmer 5 cm on the record trays), so the bottom of the unit reads as one sideboard, with covers on show in each.
- **TV:** on LG's own stand (base 47 × 23 cm), set towards the back of the worktop. The bottom of the screen is about 5.4 cm above the worktop (about 81.4 cm), its centre at about 116.5 cm and its top just under the record ledge. The C6's HDMI ports are on the left as you face it, the same side as the Series X and the PC's cable.
- **Soundbar:** about 92 × 9 × 6 cm at the front edge, 8 cm in front of the TV stand. From the sofa you look down over it, so it stays below the picture, and the TV's remote sensor still works.
- **Series X:** upright on its fan stand at the left end of the worktop, about 15 cm from the wall and the TV. Console and stand stand about 37 cm tall (to 112.4 cm), with open air above.
- **Turntable:** on a 45 cm-deep oak shelf with its top at 113 cm in the right-hand gap, big enough for a deck as large as the Audio-Technica AT-LP120X (45 × 35 cm). It is carried by a wedged through-tenon in the right upright and a hidden cleat on the right-hand wall, clear of the unit below. Keep about 40 cm clear above the deck: the lid opens to about 165 cm.
- **LP browser:** on the worktop under the turntable shelf, 33 cm clear (worktop 76 to the shelf's underside at 109). A 31.4 cm sleeve leaning back 10° stands about 31.7 cm on the browser's base, leaving 1.3 cm. At 44 × 24 cm, clear of the right upright, it holds about 50 LPs face-out, about 13 kg, which the worktop carries over the sub bay.
- **Sub bay:** open from the floor. The sub stands on a 5 cm riser (its top at 34.4 cm) under a kit shelf with its top at 38.8 cm. The shelf holds the preamp, spare stylus, cleaning kit and brush on the left and about 70 singles spines-out on the right, with 26.8 cm clear up to the hidden drawer.
- **Hidden slim drawer:** under the worktop in the sub bay, an oak front 44 × 7.4 cm (65.6–73 cm) with a finger pull underneath, on a ply box 40 × 28 × 6 cm and 250 mm runners. Remotes, batteries, styli, spare cables.
- **PC:** on an oak wall shelf about 65 × 32 cm, top at 110 cm, on the left wall about 1.3–2 m forward of the alcove, glass to the room, the back ports towards the sofa behind an oak frame with a woven cane panel, keyboard tray underneath. See "The PC shelf".
- **High shelf:** solid oak, 30 × 4 cm in three lengths housed into the uprights, top at 192 cm, wall to wall, with 28 cm to the ceiling: books up to 26 cm, a box for cables, a print and Deco B. The left shelf sits at 156 cm and the record ledge at 155.4 cm.
- **Deco B, powered:** on the right end of the high shelf. The Cat6 from the top floor comes in through the ceiling in the corner; its 12 V lead and the patch lead down to Switch 2 run along the back of the shelf into a groove routed in the back of the right upright, down to the cable slot and on to an always-on socket of the 6-way. Its light shows solid green once it's online.

The artisan oak wall: every console fits, every record faces out

Front view to scale. Oak uprights and pegged shelves above; bays 1–2 on a 9 cm plinth, a ply bay with two face-out record trays, open sub bay below. Heights

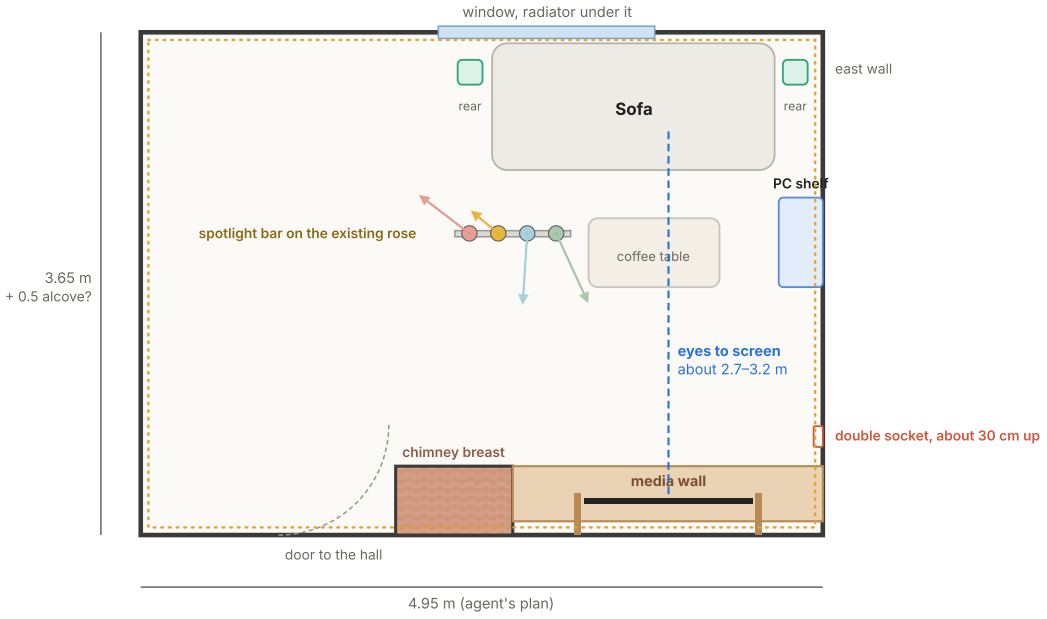


225 cm wide · solid oak worktop 225 × 40 × 3, top at 76 · unit 10 cm off the wall, 10 cm cable slot behind the worktop

Gaming Vinyl Sound optional, or keep clear

The living room from above, with the new layout

To scale. Sofa opposite the TV against the window wall; the two back spots light the books and records, the two front ones the room, none at the screen



From the sofa: with the sofa against the window wall, your eyes are about 2.7–3.2 m from the screen, depending on whether the agent's 3.65 m runs to the chimney breast or to the back of the alcove. At that distance the TV's centre at 116.5 cm is about 2° above a seated eye line, which is comfortable. A 55-inch screen fills about 22–25° of your view: fine for TV and games, on the small side for films, and the most that fits between the oak uprights (a 65-inch is about 145 cm wide, the gap is 126.5). The sofa will sit in front of the radiator under the window, so leave a 10 cm gap behind it.

Why the unit is 76 cm tall

Two face-out drawers stacked need more height than a BESTÅ frame has inside, and the worktop has to stay level across the alcove because the TV spans bays 1 to 3. So the whole unit goes up 9 cm, from 67 to about 76 cm.

One face-out tier	cm
Gap under the box	0.5
Drawer base	1.8
12-inch sleeve leaning back 10° (31.4 × cos 10°)	30.9
Finger room above	1.5
One tier · two tiers · carcass with 18 mm top and bottom	**34.7 · 69.4 · 73.0**

The trade-off: the TV's centre is about 116.5 cm instead of 108, about 2° higher from the sofa. That still sits close to seated eye height, the soundbar in front stops it going any lower even on a wall mount, and in return every record in the wall is face-out (about 230 LPs and 310 singles). Once the floor plan gives the sofa distance, the plan will confirm the viewing angle.

Fit check, sized for the largest version of each console

Zone	What goes there	Space (W × D × H, cm)	Largest version (cm)	Spare
Worktop, left	Series X on its fan stand	51 wide, open above	20.5 × 20.5 footprint, 37 tall	About 15 cm each side
Worktop, centre	TV on LG's stand at the back, soundbar at the front edge	122 wide, 40 deep	Stand base 47 × 23; soundbar 92 × 9 × 6	8 cm spare in depth
Worktop, right	LP browser under the turntable shelf	44 × 24 × 33	LP sleeve 31.4, about 31.7 tall leaning on the base	1.3 cm above; about 50 LPs
Oak shelf, right	Turntable, shelf top at 113	45 × 45, 40 clear above	AT-LP120X: 45 × 35 × 14	Just fits; most decks are 42–43 wide
Bay 1, top row	Xbox One and its power brick	56 × 40 × 16	One original: 33.3 × 27.4 × 7.9	8.1 above; 23 beside for the brick
Bay 1, middle	Xbox 360 and its power brick	56 × 40 × 16	360 original: 30.9 × 25.8 × 8.3	7.7 above; 25 beside for the brick
Bay 1, bottom	Roll-out game drawer	56 × 40 × 25.2	PS2/360 case 19 × 13.5 × 1.4; leaning in a bin about 18.5 tall	About 3 cm above the bins; about 48 games
Bay 2, top row	PS3 and a controller basket	56 × 40 × 23.7	PS3 original: 32.5 × 27.4 × 9.8	13.9 above; 24 beside

Bay 2, middle	PS2, HDMI switch at the front edge	56 × 40 × 16.5	PS2 original: 30.2 × 18.3 × 7.8	8.7 above; 26 beside
Bay 2, bottom	PS1 and RetroTINK	56 × 40 × 17	PS1 27 × 18.8 × 6; RetroTINK up to 23 × 23 × 8	9 above; 6 beside
Bay 3, upper tray	LPs face-out leaning 10°, 7-inch singles face-out beside them	56.4 × 40 × 34.2 (37.0 to 71.2)	LP sleeve 31.4, 30.9 tall leaning; single 18.3, 18.0 tall	2.1 cm above; about 90 LPs + 120 singles
Bay 3, lower tray	The same	56.4 × 40 × 35.2 (1.8 to 37.0)	The same	2.6 cm to the upper tray; about 90 LPs + 120 singles
Sub bay, upper	Kit shelf: preamp, stylus, cleaning kit, brush, spare inner sleeves; singles spines-out	44 × 30, 26.8 high	7-inch sleeve 18.3; cleaning kit about 12	8.5 cm above the singles; about 70 singles
Sub bay, lower	Subwoofer on a 5 cm riser, facing the sofa, open in front	45 × 50 × 37	AX5100G sub 21.4 × 30.4 × 29.4	2.6 cm above; 13.6 cm behind for its port
Sub bay, top	Hidden slim drawer	40 × 28 × 6 box	Remotes, batteries, styli	—
High shelf	Books, a cable box, print, Deco B	225 × 30, top at 192	Books up to 26 tall	2 cm above the tallest books

The PS3 keeps the top row of bay 2 because it runs far hotter than the others, so its heat rises out under the worktop and through the open back. The games sit below the Xbox 360, so warm air rises away from them. The records have their own plywood carcass, away from the PS3, and stand on drawer bases on runners screwed into 18 mm ply, never on a BESTÅ shelf (20 kg limit).

Console sizes, every version

The bays are sized for the largest version of each console. The model number is on the label underneath; a smaller version just leaves more room.

Console	Version	W × D × H (cm)	Power
PS1	Original (largest)	27.0 × 18.8 × 6.0	Built in
PS1	PS one	19.3 × 14.4 × 3.8	External adapter
PS2	Original (largest)	30.2 × 18.3 × 7.8	Built in
PS2	Slim	23.1 × 15.2 × 2.8	External brick
PS3	Original (largest)	32.5 × 27.4 × 9.8	Built in
PS3	Slim	29.0 × 29.0 × 6.5	Built in
PS3	Super Slim	29.0 × 23.0 × 6.0	Built in
Xbox 360	Original (largest)	30.9 × 25.8 × 8.3	Large external brick
Xbox 360	S	about 27.0 × 26.4 × 7.5	External brick
Xbox 360	E	about 25.4 × 26.4 × 6.7	External brick
Xbox One	Original (largest)	33.3 × 27.4 × 7.9	External brick
Xbox One	S	29.5 × 22.9 × 6.4	Built in
Xbox One	X	30.0 × 24.0 × 6.0	Built in

Xbox Series X	Standing	15.1 × 15.1 footprint, 30.1 tall	Built in

The oak joinery

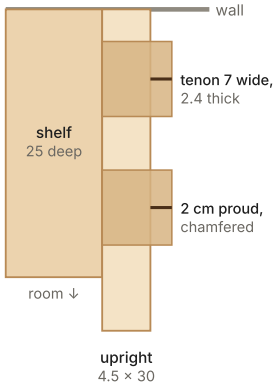
Above the worktop the wall is furniture rather than cladding: two solid oak uprights, shelves carried on joints you can see, and the Card Room Green wall left to show between them. Slatted panels are the look of the moment and will date with it; joinery like this won't.

- **Two oak uprights frame the TV:** solid oak, 4.5 × 30 cm and about 144 cm long, from the worktop (76) to the ceiling, about 2 cm clear of each side of the screen. They stand on the worktop on two dowels each, are scribed to the ceiling and held to the solid back wall with concealed brackets.
- **Wedged through-tenons:** the left shelf (top 156) and the turntable shelf (top 113) pass through the uprights on two tenons each and show 2 cm on the TV side, held with oak wedges. They are the joints you see first, so they're the ones worth cutting by hand.
- **Housed and pegged:** the high shelf, in three lengths at 192, and the record ledge are housed into the uprights and pinned with 10 mm oak pegs. The pegs show as dark dots on the uprights' faces.
- **Record ledge above the TV:** 7 cm deep with a 1.5 cm lip, top at 155.4, holding four LPs face-out against the wall up to about 186.5, just under the high shelf. The rotating "now playing" wall.
- **LED grooves:** a routed groove in each upright's inner face holds a 2700K strip that washes the green wall either side of the TV.
- **Cane screen for the PC:** an oak frame with a woven cane panel instead of slats; it hides the ports and lets the exhaust breathe.
- **Solid oak where you touch it:** a 3 cm solid oak worktop with a chamfered edge (top at about 76), the face frame in oak rather than painted, so the base reads as one sideboard, and drawer fronts with hand-cut finger pulls.
- **The green wall stays visible** behind the TV and between the shelves, so the oak reads as furniture against the colour, not a covering.
- **Notch for the ceiling strip:** a 2 × 1.5 cm notch in the back top corner of each upright lets the existing LED strip run straight past.
- **Finish:** hardwax oil throughout. A lightly limed or smoked oak for the uprights keeps away from the bright orange "new oak" look.
- **Hidden cables:** a 2 × 1.5 cm groove routed down the back of the right-hand upright carries Deco B's 12 V lead and its patch lead to Switch 2, from the high shelf to the cable slot. The TV's leads drop straight into the slot.

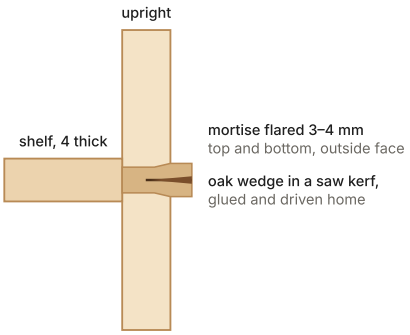
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenon, from above

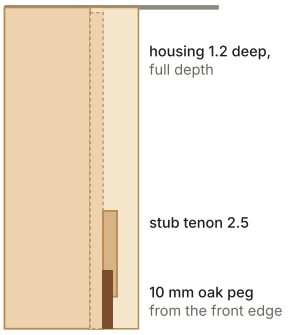


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged



high shelf / ledge
seen from above

- Through-tenons (A, B): the left shelf (top 156) and the turntable shelf (top 113) pass through the uprights and show 2 cm on the TV side.
- Housed and pegged (C): the three lengths of the high shelf (top 192) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.
- Uprights stand on the worktop on two 10 mm dowels each, scribed to the ceiling, with two concealed brackets each into the solid back wall.

Joinery cut list

Part	Material	Finished size (cm)	Joints	Qty
Uprights	Solid oak, quarter-sawn if you can	144 × 30 × 4.5 (buy 150 and scribe)	2 mortises each for the through-tenons, housings for the high shelf and ledge, LED groove inside, cable groove at the back of the right one	2
High shelf, centre	Solid oak	128.9 × 30 × 4	Housed 1.2 into both uprights, stub tenons, pegged	1
High shelf, left and right ends	Solid oak	45.7 and 46.2 × 30 × 4	Housed into one upright, on a cleat on the side wall	2
Record ledge	Solid oak	128.9 × 7 × 2.4, lip 126.5 × 1.5 × 0.9	Housed and pegged	1
Left shelf	Solid oak	51 × 25 × 4 (44.5 + 4.5 through + 2 proud)	Two through-tenons 7 × 2.4, wedged; cleat on the left wall	1
Turntable shelf	Solid oak	51.5 × 45 × 4 (45 + 4.5 through + 2 proud)	Two through-tenons 7 × 2.4 within the upright's 30 cm depth, wedged; cleat on the right wall	1
Wedges	Oak, contrasting if you like	7 × 3, tapering 4 to 0.5 mm	Glued into saw kerfs	4 + spares
Pegs	10 mm oak dowel, riven if possible	5 long	Driven from the front edge, trimmed flush	6
Worktop	Solid oak, full stave or wide board	225 × 40 × 3	Chamfered front edge; cut edges sealed	1
Face frame	Oak, 20 mm	Stiles about 5 wide, rails 2.6 and 2 tall, 225 long	Glued and pinned; end stiles scribed	1 set

PC screen	Oak frame 2 × 3 section, pre-woven cane	26 × 48	Cane in a groove with a spline	1

Compared with the slat wall	Estimate (£)
Two solid oak uprights, 150 × 30 × 4.5 cm, with the joints cut	150–300
Solid oak worktop instead of laminate	150–300 more
Oak face frame instead of painted MDF	20–30 more
No slat panels or battens	230–380 less

Overall it costs about the same as the slat wall, a little more with the solid worktop, and it's a more permanent piece of furniture. If a joiner cuts the joints, add their time. Every cost is in the build tracker under "Oak joinery and lighting".

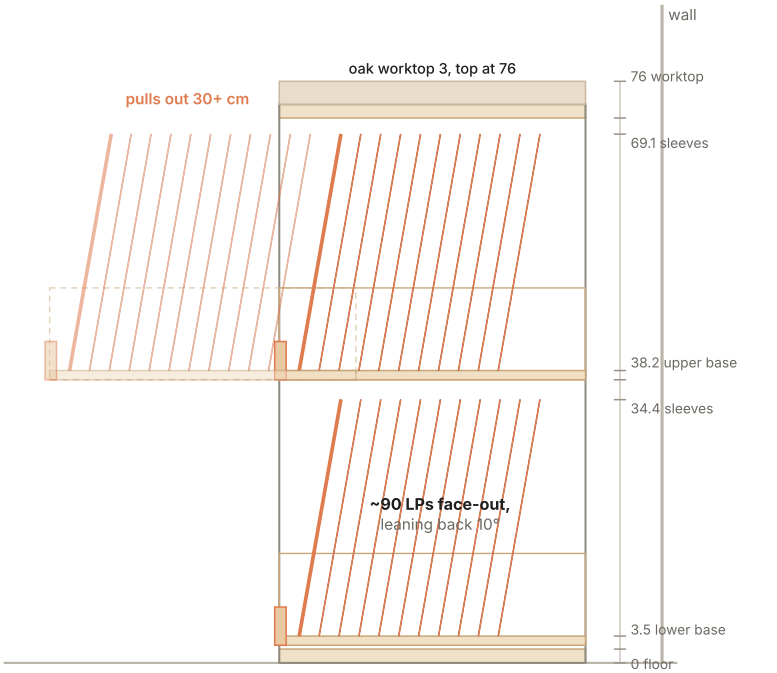
The record trays

Bay 3, under the turntable, is two roll-out trays stacked, and everything in them is face-out. LPs stand in a trough on the left of each tray, leaning back about 10° like a record shop browser; 7-inch singles stand the same way in a narrower trough on the right, where the singles and spare sleeves used to stand spine-up. With a tray shut, the room sees the front LP and the front single above a slim oak lip. Pull it out and flip through both rows of covers from the front.

The record trays: every LP and single face-out, two tiers in one bay

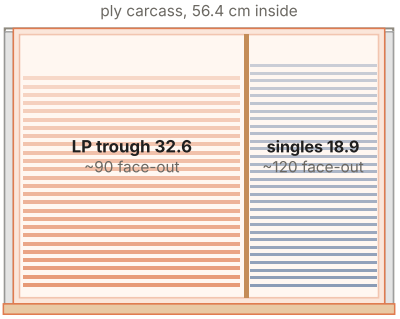
Bay 3. Section left, plan right. Pull a tray and flip through the covers like a record shop browser

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open · trays 40 deep on 400 mm runners
thinner trays: 12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Top view of one tray



tray 53.9 × 40 outside, 52.1 × 38.2 inside
9 mm sides · 6 mm oak divider · 5 cm oak lip
about 32 kg full: open one tray at a time

What changed, and why

- **Thinner trays:** 9 mm birch ply sides, front and back instead of 12 mm, a 12 mm base instead of 18 mm, a 6 mm oak divider, and a slimmer oak lip, 5 cm tall and 15 mm thick. It looks lighter from the room and shows more of each front cover.
- **What the slimmer build buys:** 4.2 cm more width inside each tray, enough to turn the singles to face-out (a 7-inch sleeve is 18.3 cm), and 6 mm more finger room above every row.
- **Spare inner sleeves** move to the kit shelf or the hidden drawer.

How the system works

- **LP trough:** 32.6 cm wide, so a bare 31.4 cm sleeve has 6 mm each side. Records lean back 10° against each other; the last one leans on a raked oak rest at the back.
- **Singles trough:** 18.9 cm wide, the same lean, a smaller rest at the back.
- **Capacity:** the inside is 38.2 cm deep. The lean takes about 5.5 cm for LPs, leaving about 31 cm of stack: about 90 LPs at about 3.3 mm each, allowing for gatefolds. Singles lean less far, leaving about 34 cm: about 120 at about 2.8 mm. Two trays: about 180 LPs and 240 singles.
- **Outer plastic sleeves:** if most of your LPs live in thick outer sleeves (up to about 32.3 cm), widen the LP trough to 33.4 cm; the singles trough then drops to 18.1 cm, too narrow for face-out singles, so they go

back to spine-up.

- **Runners:** 400 mm full-extension ball-bearing runners rated 68–70 kg per pair (45 kg is the minimum), screwed straight into the carcass's 18 mm ply sides. The whole tray clears the unit, so the back records are as easy to reach as the front.
- **Weight and safety:** about 32 kg per full tray. The carcass must be fixed to the wall with anti-tip brackets before the trays are loaded, and only one tray opened at a time.

Sizes

Dimension	Figure	How it adds up
Carcass	60 × 40 × 73 outside, 56.4 × 69.4 inside	18 mm birch ply sides, top and bottom; back open to the cable channel
Tray width	53.9 outside, 52.1 inside	56.4 opening − 2 × 1.25 runners; 9 mm sides
Tray depth	40 outside, 38.2 inside	9 mm front and back panels
Inside layout	32.6 + 0.6 + 18.9	LP trough, oak divider, singles trough
Lower tray	Underside 2.3, base top 3.5, sleeves to 34.4	Lip from 2.3 to 7.3; 2.6 cm finger room under the upper tray
Upper tray	Underside 37.0, base top 38.2, sleeves to 69.1	Lip from 37.0 to 42.0; 2.1 cm finger room under the top panel at 71.2
Capacity	About 90 LPs and 120 singles per tray	About 31 cm of LP stack ÷ 3.3 mm; about 34 cm of singles ÷ 2.8 mm
Weight	About 32 kg per tray, full	90 × about 0.23 kg + 120 × about 0.045 kg + about 5 kg of tray

Cut list

Part	Material	Size (cm)	Qty
Carcass sides	18 mm birch ply	73 × 40	2
Carcass top and bottom (between the sides)	18 mm birch ply	56.4 × 40	2
Carcass back rails (top and bottom, back open between)	18 mm birch ply	56.4 × 8	2
Tray bases	12 mm birch ply	53.9 × 40	2
Tray sides	9 mm birch ply	40 × 12	4
Tray backs	9 mm birch ply	52.1 × 12	2
Tray front panels (low, so the covers show)	9 mm birch ply	52.1 × 4.5	2
Dividers	6 mm oak	38.2 × 10	2
Raked back rests (10°)	20 mm oak offcut	32.6 × 12 and 18.9 × 8	2 + 2
Oak lips (fronts)	15 mm oak	56.8 × 5	2
Felt or cork liner	2 mm	52 × 38	2
Runners	Full extension, side mount, 12.5 mm, 68–70 kg pair	400 mm	2 pairs

The 12 mm base carries about 27 kg of records. Glue and screw it under the sides, front and back all round, and glue the oak divider down to it: the divider stiffens the middle, so the base doesn't sag over time.

How to build them

1. **Build the carcass:** glue and screw the top and bottom between the sides, square, with the two back rails. Have the merchant cut the ply to the list.
2. **Set it in place** beside bay 2 on a flat floor, shim it level with the frames' tops at 73, bolt it to bay 2's frame, and fix it to the wall with anti-tip brackets.
3. **Fit the runners** straight onto the ply sides, level, every screw hole used: the lower pair so the tray's underside sits at 2.3 cm, the upper pair at 37.0 cm. Make the tray width exactly what the runner instructions specify (normally the opening less 25 mm).
4. **Build the trays:** sides, back and low front panel glued and pinned to each other, then glued and screwed down onto the 12 mm base. Glue the oak divider 32.6 cm in from the left inside face, fit the two raked rests, and glue the felt down.
5. **Hang the trays** and check they run true empty.
6. **Fit the oak lips** from inside the tray, with a hand-cut finger pull at the top centre, 2–3 mm clear of the face frame.
7. **Load gradually,** one tray at a time, and check both still run smoothly full.

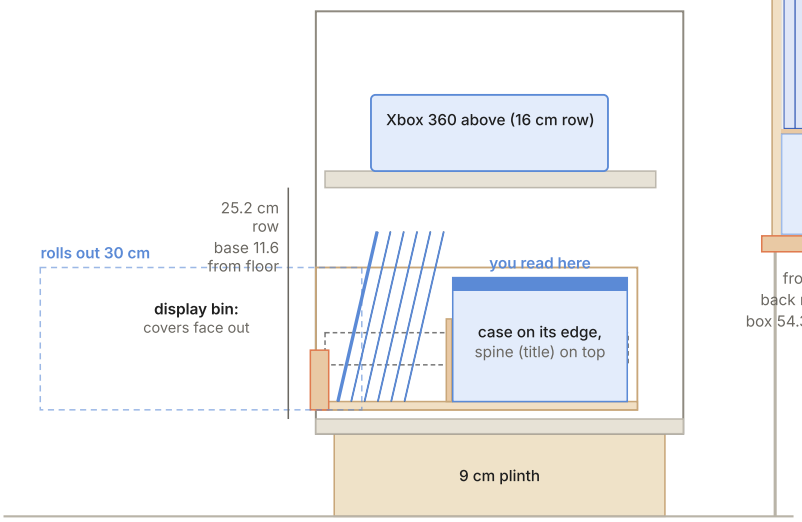
The game drawer

The bottom row of bay 1 becomes a roll-out drawer for Xbox 360, Xbox One and PS2 games. At the front, two display bins hold games face-out, leaning back slightly like the record trays, so the front cover of each is on show even with the drawer shut: Xbox on the left, PS2 on the right, with a small tray between them. Behind them, the rest of the collection stands on its long edge with the spine on top. Roll the drawer out and every title faces up at you, in rows you can read without moving anything, and any game lifts straight out.

The game drawer: covers on show at the front, every title readable from above at the back

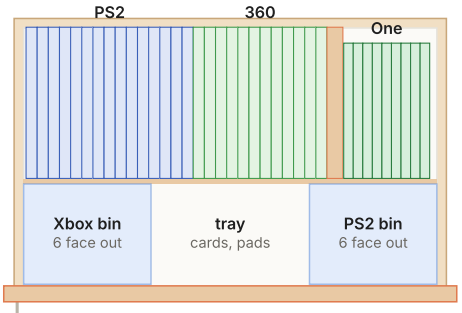
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 10 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out



front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

The cases it's built for

Console	Case (height × width × thickness)	Spine-up, it stands	Where it goes
PS2 (PAL)	19 × 13.5 × 1.4 cm (DVD-size keep case)	13.5 tall, 19 long	Back row left; PS2 display bin
Xbox 360	19 × 13.5 × 1.4 cm, same as PS2	13.5 tall, 19 long	Back row centre; Xbox display bin
Xbox One	17 × 13.5 × 1.1–1.2 cm	13.5 tall, 17 long	Back row right, behind the follower block; Xbox display bin
PS1 (PAL)	14.3 × 13 × 1.74 cm, much fatter	—	Not in this unit

Measure about ten of your own games of each kind before cutting: cases vary by 1–3 mm between publishers and replacement cases, and steelbooks are slightly bigger.

Sizes and capacity

- **Box:** about 54.3 cm wide and 35 cm deep outside, on 350 mm runners screwed to the BESTÅ sides (the same way IKEA's own BESTÅ drawers fix). With 12 mm ply sides that leaves about 51.9 cm inside. Sides

about 15 cm tall, with a 9 mm base lined in felt.

- **Front (13 cm deep):** two bins 16 cm wide, each taking one column of cases leaning back 13°, about six per bin, so 12 games are on display and you flip through them from the front. A removable 2 cm block in the Xbox bin lifts the shorter Xbox One cases level with the 360 ones. The 18.7 cm tray between them holds memory cards, a pad or manuals.
- **Back (about 19.5 cm deep):** one transverse row of cases standing on their opening edge, spine up, each 19 cm case running front to back. About 37 PS2 or 360 games fit across at 1.4 cm each, or about 43 Xbox One at 1.2 cm. A sliding follower block keeps a part-full row upright and divides the consoles. As drawn: 15 PS2, 12 Xbox 360 and 9 Xbox One.
- **In total** about 48 games, roughly 5–6 kg with the drawer about 10 kg. That's fewer than the 80 the old two-deep upright row held, but none are hidden.
- **Height:** a 25.2 cm row. Base clearance, the drawer base and a leaning 19 cm case come to about 22 cm, leaving about 3 cm to get fingers in.
- **Is spine-up safe?** Yes. The disc still stands vertically, as in book-style shelving, which archival guidance prefers over lying flat. Don't pack the row so tight the cases bow, and keep the covers out of direct sun.

Runners

Runner, 350 mm	Type	Rating	Note
Accuride DZ3832EC (premium)	Full extension, soft-close, side-mount	43 kg	About 30 cm of travel exposes the whole back row
Screwfix Essentials 964VN (value)	Full extension, self-closing, ball-bearing	40 kg	Easy to buy locally
IKEA BESTÅ drawer runners	Soft-close or push-open	—	Not recommended: reported as not full extension, and push-open fires from a stray foot

At about 10 kg, any of the first two has at least four times the load it needs.

Cut list

Part	Material	Size (cm)	Qty
Box sides	12 mm birch ply	35 × 15.5	2
Box back and front panel	12 mm birch ply	51.9 × 15.5 (back), 51.9 × 6 (front)	1 + 1
Base, in a 6 mm groove	9 mm birch ply	52.5 × 34.4	1
Cross divider	6 mm ply	51.9 × 9	1
Bin walls	6 mm ply	12.6 × 9	2
Bin floors (raked)	6 mm ply	16 × 12.6	2
Follower blocks	20 mm oak offcut, felted	19 × 12	2
Xbox One lift block	20 mm offcut	15 × 12	1
Oak lip (front)	20 mm oak	56.8 × 6.5	1
Runners	Full extension, soft-close, side mount	350 mm	1 pair

How to build it

1. **Fix the frames to the wall first**, on their plinth, as for the record trays.

2. **Fit the runners** to the frame sides, level, with the drawer base about 1 cm above the frame's bottom panel; drill pilot holes, as BESTÅ sides are particleboard. Use the frame's pre-drilled holes or short chipboard screws; if a screw won't hold, add a 12 mm ply packer.
3. **Build the box** from 12 mm birch ply: sides and back about 15 cm tall, a 9 mm base in a groove, glued and screwed. Make its width exactly what the runners specify.
4. **Add the dividers:** a cross divider 13 cm back from the front, and two bin walls 16 cm in from each side. Fit the two bin floors with a slight back-rake so cases lean about 13°.
5. **Make the follower block** from a 2 cm offcut, 12 cm tall and 19 cm long, felted underneath so it slides; one or two of them separate the consoles.
6. **Fit the front:** a 6.5 cm oak lip with a hand-cut finger pull, in the same oak as the record trays, overlaying the face frame with a 2–3 mm gap.

Options

- **PS1 games:** stored outside this unit. PAL cases are about 14.3 × 13 × 1.74 cm, about 57 per metre of shelf.
- **More games:** past about 48, keep the overflow spine-up in a high-shelf box, rotating favourites into the drawer.
- **Sorting:** label the follower blocks by console, and keep the bins for what you're playing now.

Storage: every record face-out

With PS1 games out of the unit, the vinyl side is set up so every LP and single shows its cover.

Where	What	How many	Notes
Bay 3, upper tray	LPs and singles, face-out	About 90 LPs + 120 singles	Front covers on show above the lip
Bay 3, lower tray	LPs and singles, face-out	About 90 LPs + 120 singles	The same, at floor level
Under the turntable	Oak LP browser, face-out	About 50 LPs	Shelf at 113 leaves 33 cm clear; the front record is on show beside the TV
Above the TV	Oak record ledge	4 LPs	The "now playing" wall
Sub bay, kit shelf	Preamp, stylus, cleaning kit, brush, inner sleeves; singles spines-out	About 70 singles	Shelf top at 38.8, over the sub
Sub bay, top	Hidden slim drawer	—	Remotes, cables, batteries, styli
High shelf	Books, a box for cables and controllers	About 40–50 books	Books up to 26 cm; LPs are too heavy up here (about 70 kg per metre)
Total in the wall		**About 230 LPs, all face-out, and about 310 singles**	240 of the singles face-out

If the collection grows

- **Overflow cube unit:** a 1 × 4 or 2 × 4 cube unit such as IKEA KALLAX (cubes about 33 cm inside, 39 cm deep) holds about 60–70 LPs per cube. On its side behind the sofa or under a window, a 2 × 4 holds about 500 LPs and gives you a second display top.
- **Tighter packing:** each tray takes about 95 LPs packed tight, at the cost of an easy flip.
- **LP browser depth:** keep it face-out at 24 cm deep. A deeper rack under the turntable would weigh about 35 kg over the sub bay, more than the worktop should carry there.

Paint: Farrow & Ball Card Room Green

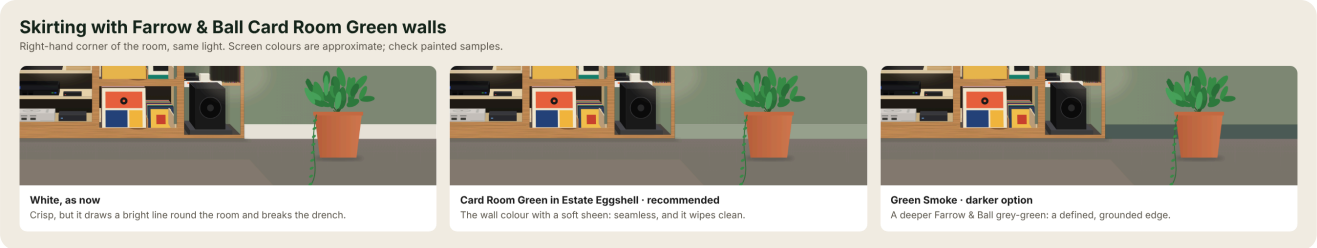
The room is colour-drenched in Farrow & Ball Card Room Green No. 79, your pick from the last grid: walls, ceiling, alcove, skirting, door and frame, radiator and the PC trunking, all one colour. It's a soft, neutral grey-green, calm without feeling cold, a little lighter and greyer than Sage Green. The oak, the records and the colours of the spotlight bar do the contrast work.



The artisan oak version in Card Room Green (top left, chosen) beside Sage Green, Treron and the earlier deep green. Screen colours are approximate and the renders are lit as in the evening, so judge from painted samples.

- **Finishes:** Estate Emulsion (chalky, flat) or Modern Emulsion (more wipeable) on walls, ceiling and alcove; Estate Eggshell in the same colour on skirting, door, frame, radiator and trunking. Modern Eggshell is the tougher choice for the skirting if it takes knocks.
- **Ceiling:** the same colour keeps the drench seamless and makes the LED strip's wash read as one soft band.
- **For the OLED:** Card Room Green is a mid-tone (LRV about 27): matt and muted enough that the screen keeps its contrast in the evening. Turn the spots down while watching and keep the bias light behind the TV.
- **Choosing:** paint an A4 board and look at it by day and with the LED strip on: warm LEDs push greys towards green, cool daylight towards grey.
- **Order:** paint before building anything, alcove included, so nothing needs cutting in around the uprights and shelves.
- **How much:** measure walls and ceiling. A mid-tone over pale walls usually covers in two coats; over the old white gloss on the woodwork, prime first.

Skirting: paint it Card Room Green too



- **Recommended: Card Room Green in Estate Eggshell.** The wall colour with a soft sheen. It completes the drench, so the room's edges disappear and the oak base and records are what you notice, and it hides scuffs from feet and the hoover far better than white.
- **Darker option: Green Smoke.** A deeper Farrow & Ball grey-green with a hint of blue, for a defined edge that grounds the room. It draws the eye down, so keep it to the skirting and door.
- **Keeping a light trim:** swap the brilliant white for a soft off-white. Bright white next to a muted grey-green looks cold and stark.
- **Painting over white gloss or satin:** wash, key with a fine sanding pad, prime with a woodwork primer, then two coats of eggshell. Do the door, frame and window board at the same time.
- **Behind the unit:** where the base meets the side wall inside the alcove, cut the skirting back so the oak end stile sits flush, or scribe the stile over it.

Lighting

Three layers, all warm: the LED strip you already have round the ceiling, light built into the oak, and a spotlight bar on the existing ceiling rose in the middle of the room.

Now: the LED strip round the ceiling

- **What you have:** a strip stuck to the ceiling (not the wall) round the whole edge of the room, facing down, as in your photo. In the renders it runs along the left wall, into the alcove, across its back wall and reveal, and across the chimney breast, washing the top of every wall.
- **Where it meets the oak:** the uprights run up to the ceiling against the back wall, so the strip would stop at each one. Cut a small notch, about 2 × 1.5 cm, in the back top corner of each upright so the strip runs straight through.
- **Keep it warm:** 2700K, on a dimmer, so it matches the rest.

Later: upgrading the strip

Upgrade	What it does	Effort
Aluminium channel with an opal diffuser, in the same place	Hides the individual LED dots and gives a clean line; cools the strip so it lasts longer	Easy: stick or clip the channel up, re-lay the strip
LED coving or a shadow-gap cornice	Hides the strip completely and throws light up across the ceiling instead of down the walls: softer, no glare from the sofa	Medium: fixed and painted Card Room Green with the walls
Strip in the uprights' grooves only, on a separate circuit	Light the media wall on its own for films, with the room strip off	Planned already; needs the fused spur for the driver

Built into the oak

- **Upright grooves:** a 2700K strip in an aluminium channel in a groove on each upright's inner face, washing the green wall either side of the TV.
- **Bias light:** a 6500K strip on the back of the TV, from its USB port, which is easier on the eyes in a dark room.

The ceiling light: a four-head spotlight bar

On the existing ceiling rose, in the middle of the room in line with the alcove's right-hand edge, in the style of Dunelm's Elements Wolston bar: a slim white bar with four conical shades on short stems, each with a light-wood topper, in your photo's colours of sage green, pale blue, mustard and dusty pink. The colours pick up the record sleeves, the wood toppers echo the oak, and against Card Room Green the pastel shades read as soft colour rather than clutter.

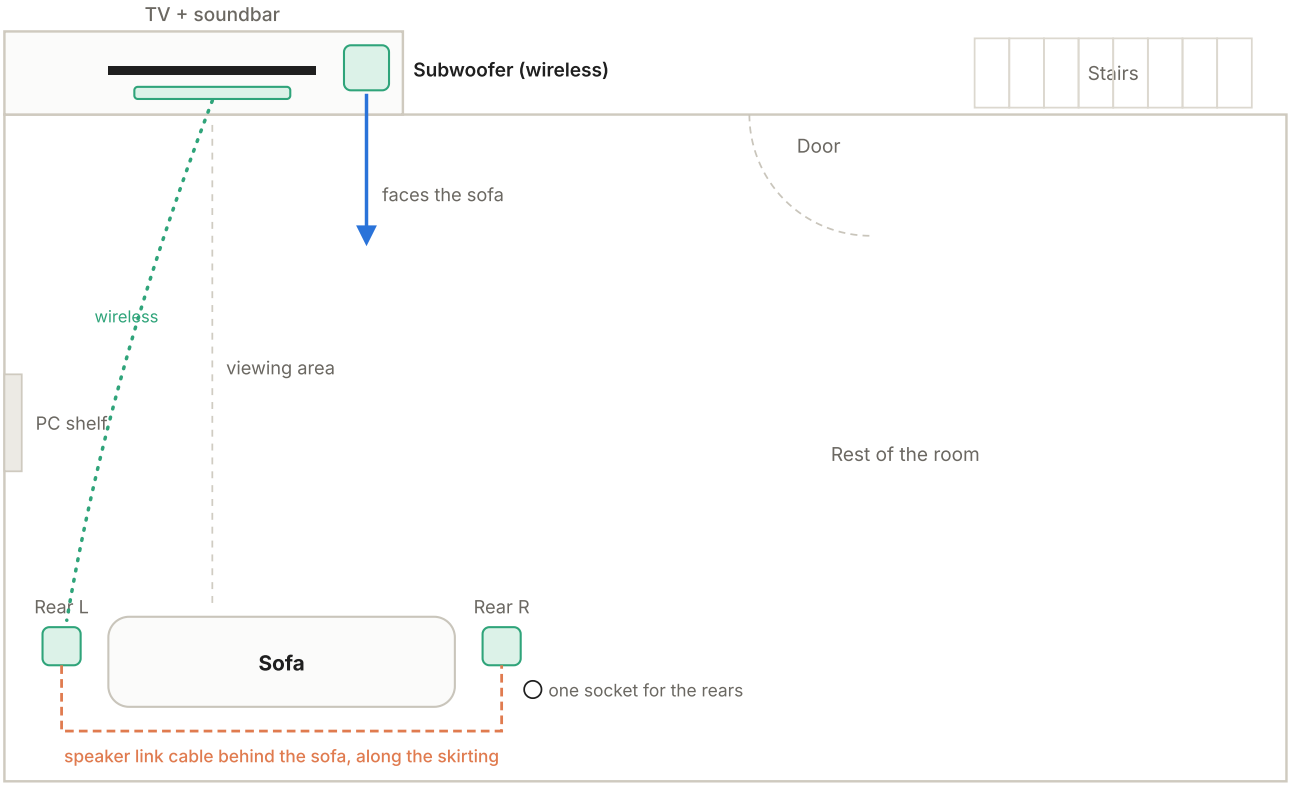
Wolston 4-light spotlight bar	Figure
Price	£49 at Dunelm
Size	84 cm long, 16 cm deep, 22.4 cm drop; shades 15 cm across
Build	Steel, adjustable heads, wood-effect toppers; semi-flush on a ceiling plate
Bulbs	4 × E14 (small screw), 6 W max each, not included; dimmable with compatible E14 LED bulbs
Rating	IP20 (indoors), Class 1: needs wiring in to the ceiling rose

- **Which way it runs:** parallel to the media wall, as drawn, so all four heads can reach their targets.
- **Aiming:** the two heads nearest the wall (sage and blue) light the books, the record ledge and the high shelf; the other two (mustard and pink) point down into the room and the coffee table. Never point a head at the TV, and tilt the front pair so the shades hide the bulbs from the sofa.
- **Bulbs:** four warm-white (2700K) dimmable E14 LED bulbs, 4–6 W each, so they match the strip and the uprights.
- **Dimmer:** swap the wall switch for an LED-compatible dimmer; turn the spots right down for films.
- **Headroom:** with a 22.4 cm drop on a 220 cm ceiling, the lowest shade sits at about 198 cm. Tilted heads hang a little lower, so if it's over the walkway, angle them up rather than down at the front. Measure the room's ceiling before buying.
- **Fitting:** it replaces the bare pendant on the existing rose; have the electrician wire it and the dimmer on the same visit.

Sound layout

The sub fires down the room at the sofa

Living room from above, not to scale. Revised: the AX5100G rears are joined by a cable, so plan one socket and a run behind the sofa



With the sofa directly opposite the TV at the left of the room, the sub fires straight at the seats, and nothing sits in the walkway by the door.

- **Subwoofer:** wireless, in the alcove's right-hand corner, facing the sofa. A corner gives the most bass, and deep bass spreads in every direction, so it blends in from there. Keep 10–15 cm between the sub and the wall behind it. A 5 cm riser lifts it off the floor so the bass stays tight.
- **Rear speakers:** one each side of the sofa, at about seated ear height or slightly above, level with or just behind your heads, angled towards the seats. On the AX5100G they're joined by a cable and only the link to the soundbar is wireless: run that cable behind the sofa along the skirting, and plan one socket for them.

- **Soundbar:** centred in front of the TV, pointing straight at the sofa.
- **Tuning:** sit on the sofa and set the sub and rear levels with the soundbar's remote or app. If the bass booms, turn the sub down rather than moving it.
- **Open room:** the sofa area is the left part of a larger space, so bass can sound a little lighter at the seats than in a closed room. The corner sub helps make up for it.

Storage without crowding

- **Inside the unit:** about 180 LPs and 240 singles face-out in the two record trays, about 70 singles and the vinyl kit over the sub, about 48 games in the game drawer (12 face-out, the rest spine-up), and a controller basket beside the PS3.
- **On the worktop and ledge:** about 50 LPs in the browser under the turntable and 4 on the ledge above the TV, so about 230 LPs in all, every one face-out.
- **High shelf:** about a metre of books between oak bookends, one lidded box for spare cables and controllers, a print and Deco B.
- **Left of the alcove:** the oak shelf through-tenoned into the left upright, top at 156 cm, above the Series X, for display or favourite games.
- **Keep clear:** the band around the TV, the 40 cm above the turntable for its lid, the air above the Series X, and the front of the sub.
- **Maximum storage:** if you need more, swap the high shelf for shallow wall cabinets across the left two-thirds, keeping the right end open for Deco B. It adds a lot of hidden space but makes the wall look heavier.

Build options

Option	How	Good	Watch out
Frames on a plinth + custom record bay + full-width top (recommended)	Two IKEA BESTÅ 60×40×64 frames on a 9 cm plinth, an 18 mm ply carcass for bay 3, and a solid oak worktop cut to 225 × 40 cm, resting on felt-lined battens over the open sub bay, all behind an oak face frame	Built-in look, mostly flat-pack, shelves on pins, room for two face-out record trays	Top at about 76 cm, so the TV sits 9 cm higher; back panels off; oil the worktop all over
Kallax 4×2 on its side	Cube unit plus side fillers	Cheapest; cubes are about 33.5 cm square inside, so LPs and single consoles fit	147 cm wide leaves a gap even after the sub bay; no room for the record drawers
Custom built-in	18 mm board carcasses built to the alcove	Exact fit throughout	Most work, and permanent

Easy use and access

- Disc trays, power buttons and front USB ports at the front edge of each bay.
- Every drawer and tray runs on full-extension runners with a finger pull in the oak lip; no push-to-open at floor level. Open one record tray at a time.
- Label each switched socket and each HDMI switch input with the same console name.
- The controller basket holds the TV and HDMI switch remotes plus a charging dock for the Xbox and PlayStation pads, powered from the USB charger.
- Keep console bays open, or use cane or mesh fronts, so air and the HDMI switch remote both get through.

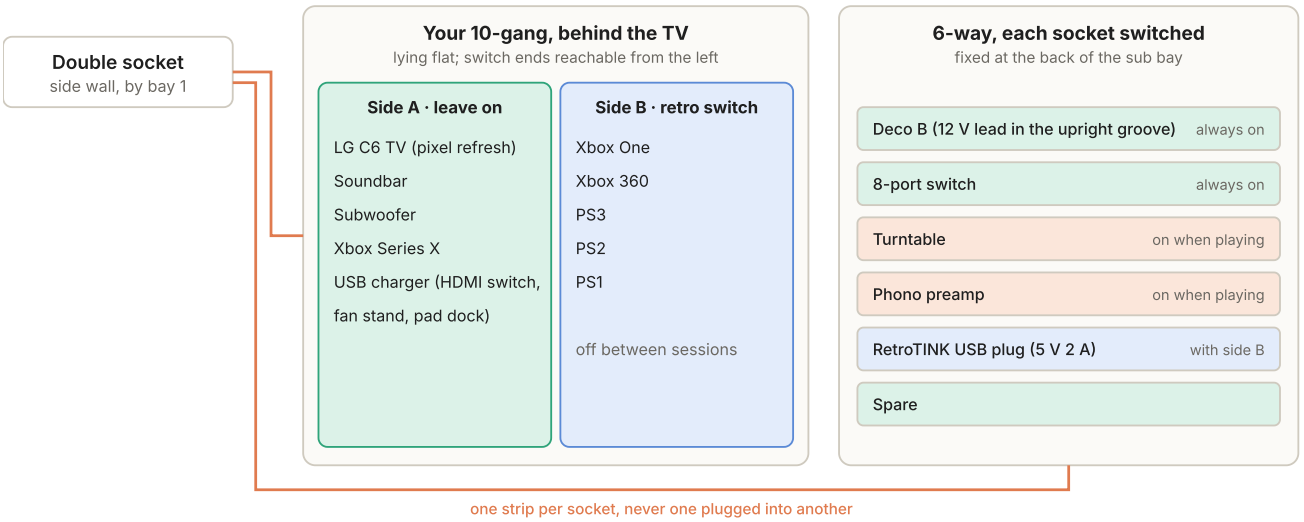
Cables, power and airflow

- **Service channel:** with the unit 10 cm off the wall, every cable drops through the 10 cm slot at the back of the worktop into two-compartment trunking: mains on one side, HDMI, Cat6 and audio on the other. Fit brush grommets at the back of each bay, and velcro-tie and label every cable.
- **To the TV:** it stands on the worktop, so its power, four HDMI leads (PC, Series X, HDMI switch, soundbar eARC) and Ethernet drop straight down behind it. No socket behind the TV is needed.
- **To Deco B:** a patch lead runs in slim corner trunking from the high shelf down to the 8-port switch. The switch is 10.1 cm deep, just too deep to lie flat in the 10 cm channel, so stand it on its edge or screw it to the back of the unit using its wall-mount slots.
- **Power bricks:** the original Xbox 360 and Xbox One bricks sit on the free half of their console's row in bay 1, where air reaches them.
- **Turntable:** its leads drop through a grommet at the back of the worktop, behind the browser, straight to the preamp on the kit shelf below; the preamp's output runs along the data side of the trunking to the soundbar's AUX input. Keep them away from the power bricks and the sub's mains lead.
- **To Deco B:** the Cat6 from the top floor comes in through the ceiling in the alcove's right corner. Its 12 V lead and the patch lead to Switch 2 run along the back of the high shelf into the groove behind the right upright, down to the slot, and to an always-on socket of the 6-way at the back of the sub bay. If the adapter's lead won't reach, add a DC extension with the same plug rather than a mains extension up at the shelf.
- **To the subwoofer:** its power lead runs along the channel to the always-on strip. Put felt or rubber pads where the worktop rests on its wall batten, so bass doesn't rattle it.
- **PC on the left wall:** HDMI and Cat6 drop through the shelf notch, down the trunking and along the skirting into the service channel. Over 3 m, use an active optical HDMI cable; these work one way round only, so the end marked "source" goes into the PC.
- **Power:** the wall socket on the side wall just in front of bay 1, swapped from a single to a double, feeds your 10-gang and the 6-way, one per socket, never one plugged into the other. You won't need to reach it day to day: the strips' own switches do that. See the power map below.
- **OLED care:** keep the TV on the always-on strip. It runs its pixel refresh while in standby, so don't switch it off at the wall.
- **Airflow:** no doors on the console bays (use mesh or slatted fronts if you want them), open backs, and at least 4 cm, ideally 10 cm, clear around the Series X.

Power map

Power map: your 10-gang plus one small switched strip

Fifteen things need power; the 10-gang takes ten, a 6-way with individual switches takes the rest



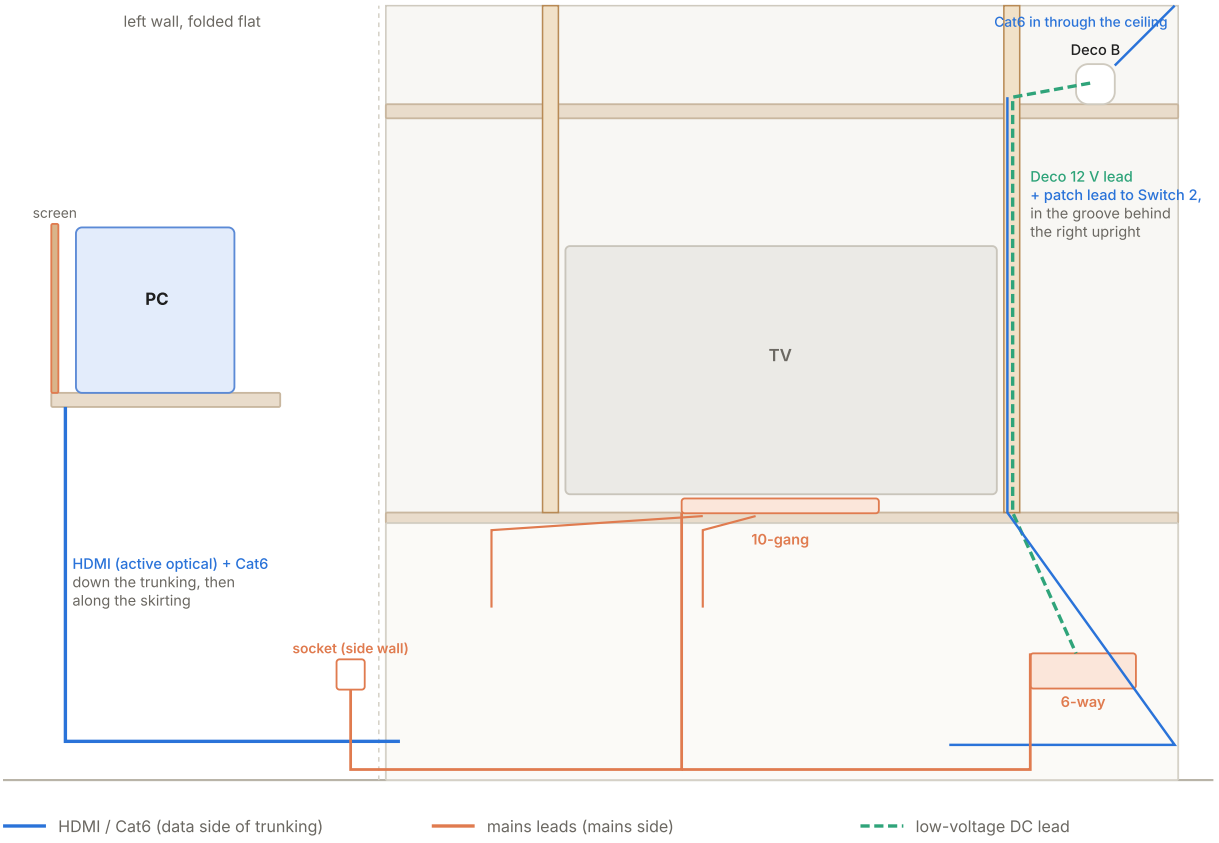
- The socket on the side wall by bay 1 is a single today: swap it for a double, one strip per socket.
- PC: its own wall socket by the PC shelf (it can draw 500–850 W); keep it off these strips.
- Rear speakers: one socket by the sofa.
- Load check: side A about 0.6 kW, side B up to about 0.8 kW with a launch PS3 running; well under the 13 A (3 kW) limit.
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles.
- Label both switch ends: “A: leave on” and “B: retro”.

- **Where the 10-gang goes:** lying flat on the worktop behind the TV, switches up, with side B's switch at the left end. From the left, reach in through the 15 cm gap between the Series X and the TV. From the sofa, the soundbar and the TV's bottom edge hide it.
- **Leads up to it:** console leads rise through the slot behind the worktop; label each plug at the strip end.
- **The 6-way:** screwed to the back of the sub bay, just under the kit shelf, so its switches are in reach and its leads are short to the deck, preamp and Deco lead.

Cable routes

Cable routes: everything ends in the channel behind the unit

Front view, schematic. The left wall is folded flat so the PC's route is visible

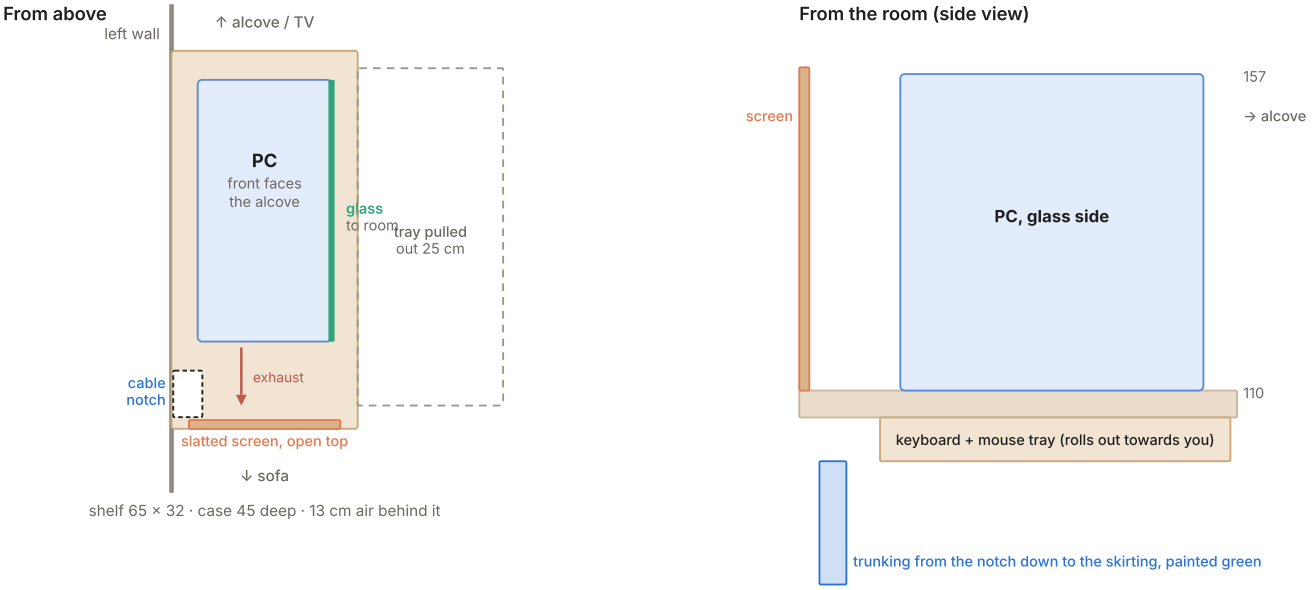


The PC shelf

A standard PC case has its glass window on its left side. With that glass facing the room, the front of the case points at the alcove and the back, with all its ports and leads, faces the sofa. The design hides that back without blocking the hot air leaving it, and parks the keyboard and mouse underneath.

The PC shelf: glass to the room, ports and cables screened from the sofa

A standard case has its window on the left, so with the glass facing the room the back ports face the sofa



- **Layout on the shelf:** the shelf runs 65 cm along the wall and 32 cm out. The 45 cm-deep case sits 5 cm from the alcove end, leaving 13 cm behind it, then an oak frame with a woven cane panel at the sofa end, 26 cm wide and 48 cm tall, the height of the case.
- **Airflow:** the screen's top is open and the cane breathes, so the rear exhaust fan's air rises out of the 13 cm gap instead of being trapped. Keep the front (alcove end) and the glass side clear for intake.
- **Cables:** a 6 × 8 cm notch in the shelf's back corner, against the wall at the sofa end, with a brush grommet. Every lead (power, HDMI, Cat6) goes from the ports into a braided sleeve, through the notch, and down slim trunking painted the wall green to the skirting. A right-angle C13 power lead keeps the plug flat to the case.
- **Power:** ideally a socket under the shelf, added on the electrician's visit, so the PC's 500–850 W stays off the media wall strips. Then only HDMI and Cat6 travel along the skirting.
- **Keyboard and mouse tray:** build the shelf as a 9 cm-deep oak box: the shelf on top, and a 7 cm housing underneath with a tray on 300 mm light-duty runners that rolls 25 cm out towards the room. It holds a full-size wireless keyboard and mouse. The housing stops 12 cm short of the sofa end so the cables can drop past it. Pull it out, lift the keyboard onto your lap, and park it again after.
- **Fixing:** hang the box on a French cleat plus two steel brackets rated 25 kg or more; the PC weighs about 10–18 kg. Into masonry, use plugs and screws; on plasterboard, fix into studs or use heavy-duty cavity fixings rated for the load.

Alternative: the slat wall

Kept for reference. The base, drawers, records and consoles are identical; above the worktop, oak acoustic slat panels on battens run from the worktop to the ceiling instead of the uprights, the shelves sit on hidden cleats, and the face frame is painted. It's quicker to fit, hides cables in the batten gap and softens sound, but it follows a current trend.



The slat wall alternative in the same Card Room Green; numbers follow its own key, which comes with the full-resolution renders.

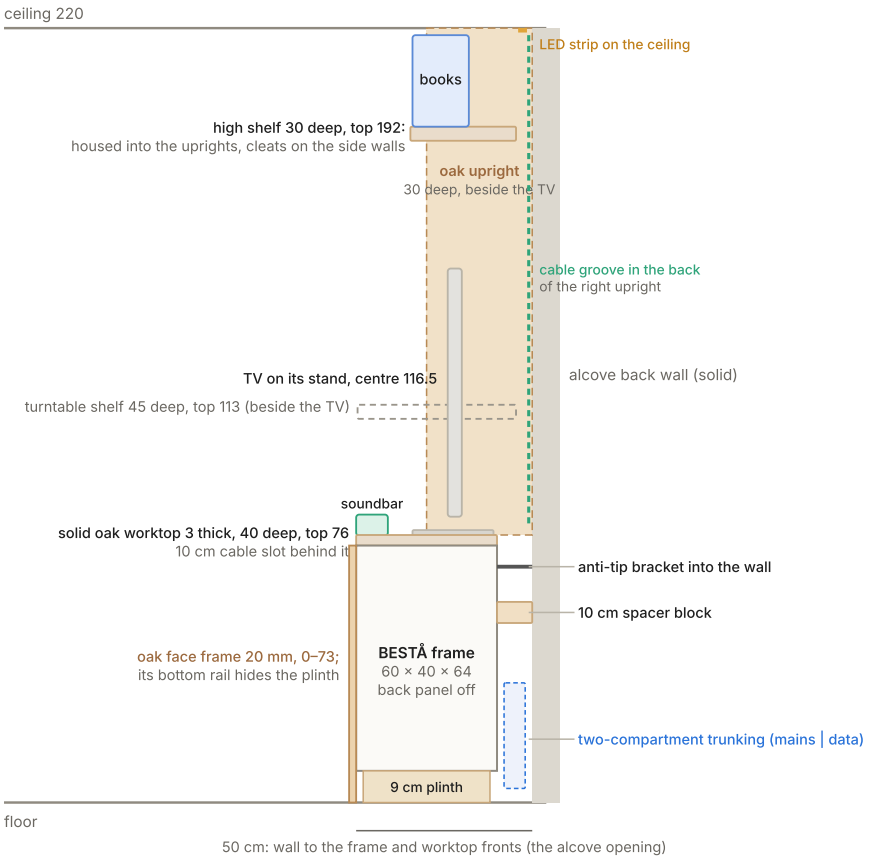
- **What changes:** four 240 × 60 cm slat panels on 25 × 50 mm battens (about £230–380), a laminate worktop instead of solid oak, an MDF face frame painted Card Room Green eggshell, and the shelves on cleats with no joints on show.
- **What stays:** every height, the record trays, the power and network routes and the lighting, except that the LED wash comes from strips under each shelf instead of the upright grooves.

Build guide: the whole wall

Everything needed to build the unit, the oak joinery, the trays and drawers, and the PC shelf. Sizes are finished sizes in centimetres. Measure your own alcove first: walls are rarely square, so take its width at the floor, at 76 cm, at 192 cm and at the ceiling, and cut the worktop and shelves to the smallest.

How the wall goes together, layer by layer

Side section through the centre of the alcove, to scale. Room on the left



What it's made of

Element	Made from	Finished size	Fixing
Plinth (bays 1–2)	18 mm ply ladder frame: front and back rails, four cross rails	120 × 38 × 9	Shimmed level on the floor; frames screwed down onto it from inside
Bays 1–2	2 × IKEA BESTÅ frame, white stained oak effect, back panels off	Each 60 × 40 × 64, tops at 73	Joined to each other, 10 cm spacer blocks to the wall, 2 anti-tip brackets each
Bay 3 carcass	18 mm birch ply, back open	60 × 40 × 73	On the floor, bolted to bay 2, 10 cm spacers and 2 anti-tip brackets
Internal shelves	4 × BESTÅ shelf (20 kg each)	56 × 36	On pins at the fit-check heights
Worktop	Solid oak, full stave or wide board, chamfered front edge	225 × 40 × 3, top at 76	Screwed up through the frame and carcass tops in slotted holes so it can move; on felt-lined battens over the sub bay

Uprights	Solid oak	144 × 30 × 4.5 (scribed)	Two 10 mm dowels into the worktop, two concealed brackets each into the back wall
Sub bay battens	44 × 22 mm planed softwood, felt-lined	40 (right wall) + 44 (back)	Into the wall, top at 73
Kit shelf	18 mm oak-veneered ply or oak	44.4 × 30	On two 30 cm battens, top at 38.8; rubber pads
Record trays	9 and 12 mm birch ply, oak dividers and lips, 400 mm runners	Trays 53.9 × 40 × 13.2	Runners straight into the bay 3 carcass (cut list above)
Game drawer	Birch ply, oak lip, 350 mm runners	Box 54.3 × 35 × 16.4	Runners on bay 1's sides (cut list above)
Hidden drawer	12 mm birch ply box, 20 mm oak front	Box 40 × 28 × 6; front 44 × 7.4 at 65.6–73	250 mm runners on a ply cheek against the bay 3 carcass and a batten on the right wall, just under the worktop
LP browser	Oak offcut base, 20 mm oak front rail and side	44 × 24, rail 4 tall, side 9 tall	Loose on the worktop, rubber feet
High shelf	Solid oak, three lengths	30 × 4, top at 192	Housed and pegged into the uprights; the end lengths also on cleats on the side walls
Record ledge	Solid oak	128.9 × 7 × 2.4, top 155.4	Housed and pegged into the uprights
Turntable shelf	Solid oak	51.5 × 45 × 4, top at 113	Wedged through-tenons in the right upright, cleat on the right wall; clear of the unit
Left shelf	Solid oak	51 × 25 × 4, top at 156	Wedged through-tenons in the left upright, cleat on the left wall
PC shelf box	27 mm oak top, 18 mm oak sides	65 × 32, box 9 deep	French cleat plus 2 steel brackets (25 kg+)
Keyboard tray	12 mm birch ply, oak lip	About 46 × 30	300 mm light-duty runners in the box
PC screen	Oak frame with a woven cane panel	26 × 48	Screwed to the shelf from below
Face frame	20 mm oak	Stiles about 5 wide, rails 2.6 and 2 tall, 225 long, 73 tall overall	Glued and pinned to the frame and carcass fronts; end stiles scribed to the walls

Tools

- Drill-driver and an SDS or hammer drill for the masonry, with masonry bits and plugs.
- Spirit level (a cross-line laser makes the shelf heights quick), tape, combination square, stud and cable detector.
- Track or circular saw for the worktop and ply (or have the timber merchant cut the ply and oak to the cut lists), jigsaw for notches.
- For the joints: a sharp set of chisels (6, 12 and 25 mm), a marking gauge, a tenon saw, a router with a straight bit for the housings and grooves, a 10 mm drill bit for the pegs, and a mallet.
- Router with a cove bit for the finger pulls (or buy lips with the pull routed), clamps, sander, wood glue.
- Dust mask, eye and ear protection.

Order of work

1. **Check and measure:** ceiling height, alcove width at four heights, the walls for plumb, the floor for level across the alcove, and your records and games.
2. **Electrician:** Cat6 runs (the one for Deco B through the ceiling in the alcove's right corner), swap the single socket in front of bay 1 for a double (or move it back into the alcove), sockets by the sofa and under the PC shelf, the fused spur for the LED drivers, and the spotlight bar and its dimmer. Mark every cable route on the wall before painting.
3. **Paint the room** in Card Room Green, skirting and door in eggshell, alcove included, and let it cure for a few days before fixing anything to it.
4. **Mark heights** on the walls with a level or laser: 73 (worktop battens), 37 (kit shelf battens), 109 (turntable shelf underside), 152 (left shelf underside), 188 (high shelf underside), and the uprights' lines at 44.5–49 and 175.5–180 cm from the left wall.
5. **Fix the cleats and battens** on the side walls for the shelf ends and the sub bay, into the masonry, level. Cut the skirting back where the base meets the walls.
6. **Build the plinth** and set it against the spacer blocks, shimmed dead level: it decides whether the worktop sits flat.
7. **Assemble the frames** with their back panels off, stand them on the plinth, join them with BESTÅ connectors and screw them down to it.
8. **Build the bay 3 carcass,** set it beside bay 2, shim it so its top is level with the frames at 73, bolt it to bay 2, and fit the anti-tip brackets on all three.
9. **Fit the worktop:** scribe the ends to the walls, oil it all over (underside too, so it stays flat), and screw up through the frame and carcass tops in slotted holes (short screws that can't come through). Felt pads on the sub-bay battens.
10. **Fit the trunking** in the channel behind the unit, mains on one side and data on the other, and run the cables in.
11. **Fit the internal shelves** in bays 1 and 2 at the fit-check heights.
12. **Build and fit the trays and drawers:** both record trays in bay 3, then the game drawer in bay 1 (steps in their sections).
13. **Fit the kit shelf and the hidden drawer** in the sub bay, and put the sub on its riser.
14. **Cut the joints and fit the oak:** dry-fit each upright with its shelves on the bench, then stand the uprights on their dowels, fix the concealed brackets, and fit the shelves, ledge, wedges and pegs (see "The oak joinery"). Lay the Deco leads in the right upright's groove before it goes up.
15. **Build the PC shelf box,** fit the keyboard tray runners, hang it on its cleat and brackets, fit the screen and the trunking down to the skirting.
16. **Make the LP browser,** fit the oak face frame, and finish all the oak with hardwax oil.
17. **Place and connect everything,** label each plug and HDMI input, and test.

Then: lighting

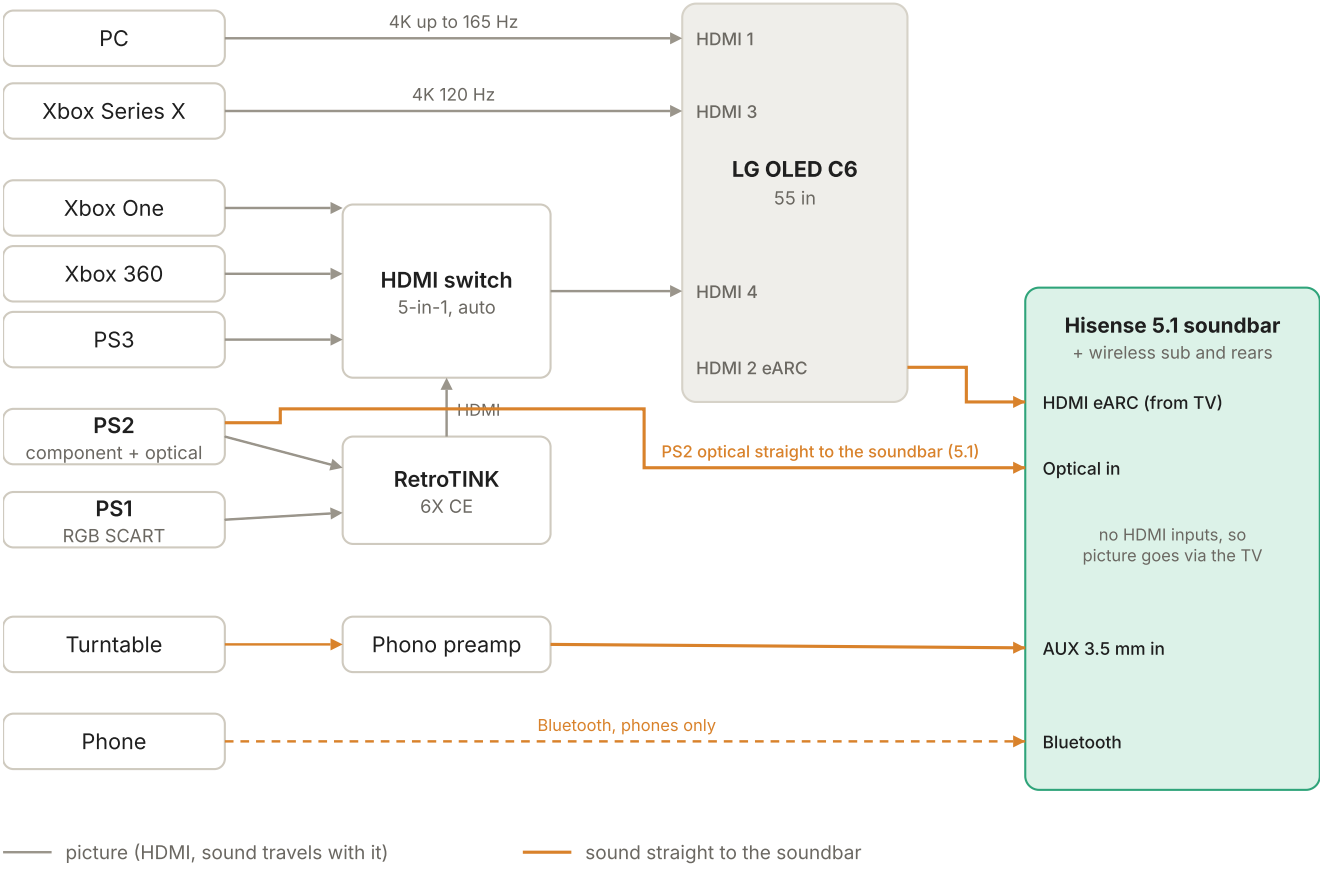
1. **Upright grooves:** aluminium channel and a 2700K strip in each upright's inner groove, wired down the back groove to a driver on the fused spur in the sub bay.
2. **Ceiling strip:** re-stick the existing strip through the notches at the top of the uprights; add the channel and diffuser whenever you upgrade it.
3. **Bias light** on the back of the TV, from its USB port.
4. **Spotlight bar:** fit it on the existing rose, parallel to the media wall, then aim the heads and set the dimmer.

Living room AV

The TV's four HDMI 2.1 ports cover everything. The PC and Series X get their own ports for 120Hz+ and VRR (the C6's 165Hz mode is for PCs), the soundbar takes eARC, and the other five consoles share HDMI 4 through a switch. The C6 has no analogue inputs, so the PS2 and PS1 go through the RetroTINK first.

Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



How each source reaches the Hisense

Source	Picture	Sound	Soundbar input
PC	TV HDMI 1	Via the TV over eARC	HDMI ARC
Xbox Series X	TV HDMI 3	Via the TV over eARC, 5.1 or Atmos	HDMI ARC
Xbox One, Xbox 360, PS3	HDMI switch → TV HDMI 4	Via the TV over eARC, 5.1	HDMI ARC
PS2	Component → RetroTINK → switch → TV	Optical, straight to the soundbar, 5.1 where the game has it	Optical
PS1	RGB SCART → RetroTINK → switch → TV	Stereo through the RetroTINK and the TV's eARC (no digital out)	HDMI ARC
Turntable	—	Preamp → AUX, straight in	AUX
Phone or tablet	—	Bluetooth, straight in	Bluetooth
TV apps	Built in	eARC	HDMI ARC

The soundbar has no HDMI inputs, so anything with an HDMI lead has to reach it through the TV; eARC carries the full 5.1. Everything that can skip the TV does: vinyl on AUX, the PS2 on optical and phones on Bluetooth. Pick the input on the soundbar's remote when you switch to the deck or the PS2; it wakes by itself when a signal arrives. If PS2 sound ever runs slightly ahead of the picture, nudge the soundbar's audio delay or AV sync setting if it has one.

TV and console settings

- Rename each TV input, with HDMI 4 as "Retro", and turn on HDMI Ultra HD Deep Colour for HDMI 1, 3 and 4.
- Use Game Optimiser on the console inputs for the lowest lag.
- Set sound output to HDMI eARC, with digital audio on Auto or Passthrough.
- Set the PS3, 360 and Xbox One to output 5.1 over HDMI.
- On the PS3, turn off "Control for HDMI" so it doesn't grab the TV input.
- For the PC at 4K above 120Hz, use a cable certified Ultra High Speed (48 Gbps), which includes the active optical one.

PS2 and retro on the 55-inch

- **Upscaler:** the RetroTINK-6X CE, $230 in the US; RetroTINK's international prices include VAT and import fees. Its motion-adaptive deinterlacing suits PS2 games, which mostly run interlaced, and it outputs up to 1440p for the TV to finish scaling to 4K.
- **Stock:** the September run sold out and the next restock is estimated for October to November, so order as soon as it's back.
- **Why not the 2X Pro:** RetroTINK now lists it as a legacy model, and it won't accept the 480p mode some PS2 games use.
- **Wiring:** component cable from the PS2 into the RetroTINK's rear RCA input for the picture, and a 2 m optical cable from the PS2 straight to the soundbar's optical input, which carries 5.1 in games that support it. The component lead's stereo plugs can still go to the RetroTINK as a fallback.
- **Power:** it comes with a USB-A to USB-C lead but no plug, and needs at least 5V 2A. Use a USB plug on its own switched socket of the 6-way, and switch it with side B.
- **PS2 settings:** set System Configuration → Component Video Out to Y Cb/Pb Cr/Pr, or you may get no picture. Turn the digital (optical) audio out on.
- **PAL games:** choose the 60 Hz mode when a game offers it at start-up, for smoother motion.
- **PS1:** an RGB SCART cable into the RetroTINK's SCART input. PS1 discs also play on the PS2 if you'd rather keep that console boxed.
- **Sharper still:** PCSX2 on the PC renders PS2 games at native 4K, with widescreen patches for many. Use your own discs and a BIOS copied from your own PS2.
- **Premium option:** the RetroTINK-4K CE ($475) outputs true 4K with the same processing, and a 6X Pro is due in the coming months. Only worth it if picture quality becomes a hobby.
- **OLED care:** 4:3 games leave black bars at the sides. Mix in full-screen viewing, and leave the TV in standby so it runs its pixel refresh.

Vinyl: through the soundbar, done properly

- **Setup:** slim turntable with a new stylus, set to PHONO → external phono preamp on the shelf over the sub → soundbar AUX input, using an RCA-to-3.5 mm lead.
- **Check first:** the external preamp only helps if the deck has a PHONO/LINE switch. With no switch, its built-in preamp can't be bypassed, so skip the external one and plug the deck straight into the soundbar AUX.

- **Why the external preamp:** a slim deck's built-in preamp is its weakest part. A proper one gives fuller bass, clearer highs and less hiss.
- **Soundbar settings:** Music mode, bass up, and surround effects, night mode and dialogue mode off. Turn the sub down slightly from your film setting if bass gets boomy.
- **No Bluetooth:** always use the AUX cable, as Bluetooth loses quality.
- **Stylus:** buy the exact replacement for the deck's model, and set the tracking weight to the cartridge maker's figure.
- **Records:** clean them with a carbon-fibre brush before playing, and keep a few spare inner sleeves in the drawer's side compartment.

Upgrade path for vinyl

Start with the soundbar route. Step up only if you want more. The old hi-fi and its built-in deck can be retired.

1. **Best for vinyl (about £200–450):** a stereo amp with phono input and sub output, the hi-fi's wired speakers, and a powered subwoofer. Fully analogue, loud and bassy. The Hisense sub can't be reused, as it only pairs with the soundbar.
2. **Best for the whole room (about £500+):** an AV receiver with phono input and HDMI eARC, replacing the soundbar and driving real 5.1 speakers. Pure stereo for vinyl, proper surround for films.

Why we chose this option

Wired where it matters, wireless everywhere else. The devices that need speed and stability are on cable, and the Wi-Fi is left free for phones.

- **Wired gaming and streaming:** the Xboxes, PS3, TV and PCs get full gigabit with the lowest latency and no dropouts, whatever the walls or neighbours do.
- **Phones get uncrowded Wi-Fi:** with heavy users on cable, phones aren't competing with a 4K stream or game download.
- **Dual-band Decos need cables to shine:** the X1500 has no dedicated band for unit-to-unit traffic. Cabling the links gives every floor the same strength as the top floor.
- **Star layout:** each Deco has its own cable from the top floor, so the kitchen doesn't depend on the living room kit being on.
- **Switches cost nothing in speed:** every port runs at full speed at once, with no setup. They're simply extra ports, about £18–25 each.
- **Zones that match the kit above them:** the Xbox bay under the Series X, the PlayStation bay under the TV, and the records under the turntable, so cables stay short and everything lives where it's used.

Options we rejected

Option	Why not
Keep the Virgin hub as router, Decos in access point mode	Loses HomeShield and parental controls
All-wireless mesh	Works, but speed drops with each wireless hop, worst in the kitchen
Powerline adapters	Slower and less stable than Cat6; fallback only if cabling is impossible
2.5Gb switches	Decos and PCs top out at 1Gb, so no gain for the extra cost
Keeping the 5-port switch upstairs	Full on day one; any new device upstairs would need another switch
Plugging devices straight into Decos	Only one spare port per Deco: not enough for 2 PCs upstairs or 6 devices downstairs
Games upright, two lines deep	The back line is hidden and you pull out the front ones to reach it
Games lying flat in stacks	Hard to pick from, and long-term flat storage can bow discs in a warm room
Records on an open shelf in bay 1	Far from the deck, spines only, and too heavy for a BESTÅ shelf; the drawer puts them under the turntable with the art on show
A green feature wall only	Leaves the alcove looking applied; the drench ties the room together
White skirting	Cuts a bright line round the room and breaks the drench; Card Room Green eggshell or a soft off-white sits better
A paper lantern for the ceiling	Replaced by the spotlight bar you chose: aimable light for the shelves, and it sits higher
A long pendant drop	On a 220 cm ceiling it would hang into the walkway; a flattened shade on a short drop keeps headroom
Keeping the deep forest green	Still a good option for the darkest picture at night, but you asked for softer; the grey-green keeps the drench and feels lighter by day
Singles spine-up beside the LPs	Thinner trays free enough width to show them face-out too, and hold twice as many

Records spine-up in one drawer	Held slightly fewer and showed only a handful of covers; two face-out drawers show every one, at the cost of a TV 9 cm higher
RetroTINK-2X Pro	Now a legacy model, won't accept 480p, and has simpler deinterlacing than the 6X CE
Cheap PS2-to-HDMI adapters	Soft, washed-out picture that a 55" OLED makes obvious
Bluetooth from the turntable	Compresses the sound; the AUX cable keeps full quality

Later ideas

- **Dave and Emma's watch log:** a shared movie and TV area, like a Spotify Blend for what you watch. Log each film or episode, mark whether you watched it together or one of you on your own, and give it a simple rating. It could live on the TV (a small web app on the LG's browser) or on your phones, and later suggest what to watch together next. First thought only; to be shaped up separately.

Still to confirm

- ☑ ~~Alcove: 225 cm wide, about 50 cm deep, floor to ceiling.~~
- ☑ ~~Every record face-out: two trays in bay 3, unit raised to 76 cm; thinner trays with singles face-out.~~
- ☑ ~~Paint: Farrow & Ball Card Room Green No. 79; skirting to match in eggshell.~~
- ☑ ~~Floor plans and room photos: alcove on the right of the hall-side wall, LED strip on the ceiling, rose in the middle of the room.~~
- ☑ ~~Main build: the artisan oak version.~~
- ☑ ~~Power: one socket on the side wall just in front of bay 1, becoming a double.~~
- ☑ ~~Lighting: LED strip round the ceiling now; one ceiling light in the middle of the room.~~
- ☐ Ceiling height: the plan assumes 220 cm (the alcove's measured height). Check it at a few points before cutting the uprights, and in the middle of the room for the pendant's headroom.
- ☐ The socket's height (drawn at about 30 cm) and its distance in front of the alcove, and whether you'd like it moved back behind the unit while the electrician is there.
- ☐ Living room depth: does the agent's 3.65 m run to the chimney breast or to the back of the alcove? And the chimney breast's width (drawn at about 85 cm).
- ☐ The brick cupboard in the kitchen: its size, and whether a Cat6 cable can come down into it from behind the media wall.
- ☐ Which room the Hub 5 is in upstairs, and the height of the low side of the sloped ceilings in the bathroom and bedroom 2.
- ☑ ~~The Xbox 360 has an HDMI port, and the black box is the Xbox One.~~
- ☑ ~~The PC connects by HDMI cable, and Cat6 runs between floors are possible.~~
- ☑ ~~The Series X uses its fan stand, on the left; the turntable is raised on the right.~~
- ☑ ~~Room: sofa directly opposite the TV at the left of the room; door about 2 m right of the alcove, with the stairs beside it; sub at the far right.~~
- ☑ ~~The alcove back wall is solid, which suits the shelf cleats and battens.~~
- ☑ ~~The TV's remote sensor works with the soundbar in front, so LG's own stand stays.~~
- ☑ ~~Kitchen cable: straight from the top floor, now that Switch 1 has spare ports.~~
- ☑ ~~Power: your 10-gang has a switch for each five-socket side.~~
- ☐ PC case: confirm the window is on its left side (as standard), so the back faces the sofa.

- ☐ 10-gang: check it's surge-protected, and where its two switches sit.
- ☐ Deco B: check its adapter lead reaches from the high shelf to the back of the sub bay (about 2.2 m); if not, a DC extension with the same plug.
- ☐ Soundbar model: the plan assumes the AX5100G (3.5 mm AUX, optical and coax inputs, wireless sub, rears linked by cable). Confirm yours.
- ☐ Rear speakers: is there a socket near the sofa, and a route behind it for the link cable?
- ☐ Left wall where the PC shelf goes: solid or plasterboard? It decides the fixings.
- ☐ Paint: Card Room Green after sampling, and the room's wall and ceiling area for quantities.
- ☐ Floor in the alcove: carpet or hard floor? The plinth and the bay 3 carcass stand on it and must sit level.
- ☐ Records: roughly how many LPs and singles, and are most LPs in thick outer sleeves? The wall holds about 230 LPs and 310 singles; outer sleeves need a wider LP trough.
- ☐ Games: roughly how many PS2, Xbox 360 and Xbox One games? Past about 48, plan overflow (see the game drawer options).
- ☑ ~~PS1 games: stored outside this unit.~~
- ☐ Turntable: a modern deck usually has a small PHONO/LINE switch on the back. Check for it, and that the deck is under 45 cm wide.
- ☐ Console model numbers (on the label underneath), to tighten the row heights. The plan assumes the largest version of each.
- ☐ RetroTINK: order on restock, then check its size against the 23 × 23 × 8 cm reserved.
- ☐ Check the PC cable route before buying the HDMI cable; it's estimated at 4–5 m, so active optical.

Sources

- Virgin Media: Hub modem mode
- RetroTINK-6X CE shop page (stock, power, box contents): retrotink.com/shop/retrotink-6x-ce
- Introducing the RetroTINK-6X CE (inputs incl. TOSLINK, deinterlacing): retrotink.com/blog/introducing-the-retrotink-6x-ce
- RetroTINK-2X Pro, listed as legacy; RetroTINK-4K CE
- Expert Reviews: LG C6 OLED review (HDMI 2.1 ×4, one eARC, ports on the left): expertreviews.co.uk/technology/tvs-home-cinema/lg-c6-oled-review
- JB Hi-Fi: LG C6 specifications (165Hz mode for PC sources)
- Hisense AX5100G spec sheet (inputs, sub size, auto power-on): assets.hisense-usa.com
- JB Hi-Fi Business: Hisense AX5100G (rears linked by cable, wireless to the soundbar)
- IKEA: BESTÅ shelf 56×36 cm, max load 20 kg; BESTÅ frame, white stained oak effect
- TP-Link Deco X1500 specifications (2 gigabit ports per unit)
- Glorious Vinyl Vault specifications (80 LPs and 24.5 kg per drawer), a check on the face-out drawer capacity
- Emuca: heavy-duty full-extension runners, 400 mm, 68 kg per pair
- Game case sizes: videogamecasesize.info; coverstitch.io/dimensions; Walvis Products (Xbox 360 14 mm, Xbox One 11 mm cases); Wikipedia: Keep case
- NIST SP 500-252 / CLIR pub121: Care and Handling of CDs and DVDs (store upright, book style)
- Screwfix: Essentials 964VN 350 mm runners; Metrol: Accuride DZ3832EC; IKEA: BESTÅ drawer runners
- Can-Am, mmdesign (DiscSox): spine-up media drawer capacities
- Sony PS3 and Microsoft Xbox 360 S manuals: ventilation and disc storage; IKEA UK: secure it
- Ideal Home: living room paint ideas (colour drenching, double drenching); The Paint Shed: colour drenching with Farrow & Ball
- Pontata: Card Room Green vs Little Greene Sage Green (approximate screen colours): pontata.com/compare/fb-card-room-green/lg-sage-green

- Dunelm: Elements Wolston 4 Light Semi Flush Spotlight Bar (£49; 84 × 16 cm, 22.4 cm drop; E14, 6 W max; adjustable heads): dunelm.com/product/wolston-4-light-spotlight-bar-1000189449
- Homebase and NC Tiles: Farrow & Ball Card Room Green No. 79 in Modern Eggshell and Estate Eggshell
- Haart Medway: floor plans and listing photos for the house (Portsmouth Close)
- Livingetc: the best sage green paint shades, according to designers (Card Room Green, Treron, Sage Green, Green Smoke)
- Wikipedia: Comparison of PlayStation consoles, Xbox One and Xbox Series X; Neowin: official Xbox 360 dimensions
- Pure Xbox: Microsoft's Series X ventilation guidance; Xbox Support on clearance
- Audio-Technica AT-LP120XUSB; TP-Link TL-SG108; ConsoleMods: PS2 Component Video Out setting